Intermittency, period adding and bifurcation curves in a language model's own inference map

Authorship to be decided by the human researcher

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Status of this draft. Results on the soft-token map without a marker are numerically validated in float64 (evidence rung 1 or 2 in the sense of Section 14); coarse maps are float32 and say so, Huginn's spectra are float32 and its accuracy runs bfloat16, and the dialogue results are sampled statistics with stated intervals. **[Planned:** Red] marks planned work and [**In progress:** orange] marks running work. Nothing here is a computer-assisted proof yet.

Abstract

What dynamical structures organize repeated inference in a pretrained language model? We study a precisely defined autonomous system: the expected next-token embedding of a frozen Qwen model is fed back at a fixed position after a fixed prefix. Its temperature–steering plane contains periodic text loops among numerically chaotic regimes, in models from 0.5B to 7B parameters. Numerical continuation links a period-adding family to the saddle-node of a primitive two-cycle. Derivatives of this fold predict controlled bottleneck passages without fitting; at two independently computed folds, 23 of 24 prospectively frozen forecasts are accurate within one iterate, failing the all-row criterion. Fold and flip curves reveal candidate shrimp organization, while superstable points through period 384 yield finite-order ratios approaching Feigenbaum's constant. In the native recurrence of the recurrent-depth model Huginn, spectral estimates at fixed points predict a Neimark–Sacker bifurcation and, in development experiments, the relaxation strengths at which convergence and chain-of-thought accuracy break; a frozen test on 200 held-out problems confirms no accuracy loss below that threshold and a collapse above it. In sampled self-dialogue, hostility steering produces dose-dependent persistence and history dependence. Replacing classifier-flagged replies before they enter the context keeps the model's own hostility, as judged by that classifier, low with a quarter of the steering dose of constant prevention. A frozen confirmation finds a small increment in advance warning from the model's in-context state, while finite Monte Carlo committor estimates detect no better warning and leave the attainable ceiling unresolved. These results connect explicit inference dynamics to testable mechanisms and practical control, without identifying the constructed map with ordinary autoregressive generation.

1 Introduction

Repeated inference with a fixed language model is an iterated map: the model's output becomes its next input. Recent work has measured chaos-like sensitivity in language-model generation [1, 2], temperature-driven changes of output statistics [3–7], periodic attractors of paraphrasing loops [8], attractor states of model-to-model conversations [9], and the stability of looped and recurrent-depth models through their Jacobians [10–14]. Feeding a probability-weighted vocabulary embedding back into a model is not new: it is used in training [15] and, without retraining, in pretrained models

[16], alongside continuous latent feedback [17]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [18].

We study one constructed but exactly defined system, the expected-embedding ("soft-token") map of a frozen, released pretrained language model, with a fixed chat prefix and a fixed position. The map is smooth, its Jacobians are available by automatic differentiation, and its periodic orbits can be computed and continued like those of any smooth map. Our contribution is the quantitative link between the map's primitive periodic orbits, their continued boundaries in a two-parameter plane, and passage times predicted from the derivatives of a periodic orbit's fold, together with the text each orbit produces. Ordinary sampled generation with a growing context is a different system; Section 9 measures how much of the mechanism survives sampling.

Contributions.

1. Phase maps of the soft-token map in temperature and steering strength, in float64, showing chaos with embedded periodic windows (Section 4).

2. A period-adding family of primitive periodic orbits whose text adds one "is X" pair per period step, with multipliers from the monodromy matrix (Section 5).

3. The mechanism: the 2-cycle "is it" dies in a fold; its normal form, computed from derivatives, predicts bottleneck passage times; the global reinjection sets the window-law prefactor (Section 6).

4. Fold and flip curves of periodic orbits continued in the parameter plane with float64 test functions: a period-6 window whose fold curves meet in two numerical cusp candidates (tangent reversal with a sign change of the quadratic normal-form coefficient) with a flip curve inside (a shrimp candidate), and two more such candidates on a period-4 fold curve (Section 7).

5. A period-doubling sequence toward chaos in a closed region of the parameter plane: flips located in float64 through $48 \to 96$ and superstable points through period 384, whose finite-order interval ratios rise to 4.478 and 4.625 (Section 8).

6. A bridge to sampled decoding (Section 9); the same organization in Qwen2.5-1.5B, with a period-doubling route, and chaos with embedded windows in Qwen2.5-7B (Section 10); and a second system, a recurrent-depth model with a predicted and observed Neimark–Sacker bifurcation, spectral predictions of its response to relaxation (question by question), native instability for some prompts, multiple-choice accuracy that tolerates the instability until it is large, and chain-of-thought accuracy that breaks at the predicted thresholds, confirmed on held-out problems by a frozen test (Section 11).

7. Misaligned behaviour as an attractor. In sampled self-dialogue a hostility proxy shows a steep, direction-specific capture with dose-dependent persistence and hysteresis. A pre-specified lead-time test is negative; a small warning from the model's in-context state is confirmed on fresh dialogues, and finite Monte Carlo estimates of the committor warn no better, leaving the attainable ceiling unresolved. Counter-steering triggered after a hostile reply is less efficient than mild prevention, but replacing flagged replies before they enter the dialogue keeps the model's own hostility low at a quarter of the dose. In Qwen2.5-1.5B the same ordering holds where the dose response rises, and every counter-steer backfires beyond its peak. Steering the deterministic map along the same direction yields a hostile period-adding cascade with the intermittency mechanism of item 3, and a period-7 body with the organization of a shrimp:

fold curves ending in two nondegenerate cusps, flip curves, and a doubling cascade to chaos inside (Section 12).

Related work. Language-model generation has been studied as a dynamical system at several levels. Sensitivity to small perturbations has been measured with Lyapunov-like quantities in reasoning traces [1, 2]. Output statistics change abruptly with sampling temperature, which several studies analyse as phase transitions or criticality [3–7]. Repetition loops are a known failure of decoding [19, 20]. Successive paraphrasing converges to attractor cycles [8], and multi-turn model-to-model conversations settle into attractor states [9]. Attractors and bifurcation diagrams have been studied in recurrent language models [21]. Expected-embedding feedback is used in training [15] and, training-free, in pretrained models [16, 22], next to continuous latent reasoning [17]; here the same kind of feedback, with prefix and position fixed, defines the map we analyse. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analysed through Jacobians and fixed points [10–14, 23–28]. The closest precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [18]. Our system differs: a released pretrained language model, the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. In neural systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [29]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [30, 31] and, through period 128, in a transformer trained on logistic-map trajectories [32]. Learned networks reproduce period-doubling routes [33], and so does gradient descent itself [34]. Normal forms and continuation have been applied to recurrent neural maps [35], and shrimp-shaped windows and periodicity staircases appear in neuron models [36–38]. Activation steering along contrastive directions [39, 40] supplies our second control parameter. The dynamical-systems tools are standard: numerical continuation and normal forms [41, 42], type-I intermittency [43], shrimps and hubs [44–46], period-doubling universality [47], and Markov state models [48]. A targeted search (12 October 2026) found no prior work combining these elements for the feedback map of a pretrained language model; that is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. Let $E \in \mathbb{R}^{V\times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$) and let $z(x) \in \mathbb{R}^V$ be the next-token logits produced by the model, given a fixed chat prefix (system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and user turn "Hello!") followed by a single input embedding x. The soft-token map is

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included, and v is a contrastive steering direction ("philosophy" prompts against neutral prompts, last-token states, normalized to the mean residual norm). Positions are fixed (the window does not grow), so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces to greedy decoding of the same fixed-prefix, one-token-window system when the argmax is unique; it is not ordinary generation with a growing context.

Precision. Hugging Face RMSNorm and rotary-embedding tables compute in float32 even when the model is cast to float64; we patch both. float32 maps are accurate to 3×10^{-6}–2×10^{-5} per step,

which is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are float64.

3 Methods

Phase maps. For each cell, 300 transient and 300 measured steps; the largest Lyapunov exponent from a renormalized twin at offset 10^{-6}; the period as the smallest $q \le 32$ with return distance below 10^{-8}; the attractor's identity as its rotation-invariant leading-token loop. Maps of 256^2 cells take about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes label where two tokens tie for the maximum. We call two labelled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Periodic orbits. Newton's method on $F^p(x) - x$ with forward-mode Jacobians; multipliers are the eigenvalues of the monodromy matrix, refined by a short Arnoldi factorization with float64 Jacobian-vector products where the dense matrix is not float64.

Continuation. Pseudo-arclength continuation of periodic branches in one parameter, and of fold and flip curves in two, using minimally augmented systems [41, 42]: the test function g of the bordered system $[[M \mp I, b], [c^\top, 0]]$ is obtained by GMRES with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. A reversal of the parameter tangent flags a cusp candidate, which we test with the normal form.

Normal form and passages. For the two-step map $G = F^2$ at a fold with right and left eigenvectors r, ℓ ($\|r\| = 1$, $\ell^\top r = 1$), $u' = u + a\varepsilon + bu^2$ with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$; b is computed by central differences of float64 Jacobian-vector products over a step-size plateau (later cross-checks use nested automatic differentiation), and a includes the steering's effect on the prefix. The signed coefficients are kept in the original eigenvector gauge, with ε in the free parameter (α or $\log_{10} T$; its units apply to a). For passages we set $v = \operatorname{sign}(b)\, u$ and take the ghost side $p - p_c = \operatorname{sign}(ab)\, \epsilon$ with $\epsilon > 0$, so that $v' = v + |a|\epsilon + |b|v^2$ to quadratic order. Full and midpoint passages start at $v = -U$ and $v = 0$ and end at the first strict crossing $v > U$, with $U = 0.1/|b|$. Passage counts are compared with the iterated quadratic map and with the arctangent formula [43].

Period-doubling cascade. Along a cut in one parameter, the flip of each period-p orbit is located as the root of $\operatorname{Re} \mu_c + 1$, where μ_c is the multiplier with the smallest real part, by the Illinois variant of regula falsi with warm-started orbits. Because all but one or two multipliers are small, Newton's method on $F^p(x) - x$ uses the monodromy only on a six-dimensional dominant subspace. That subspace comes from subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed; the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted: the critical multiplier must be real and equal to -1, its Ritz residual below 10^{-6}, and the other computed Ritz values (six in all, estimates of the dominant spectrum) inside the unit circle. The period-$2p$ orbit is then reached by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

4 Chaos with periodic windows

Figure 1 shows a 256×256 float64 map over $\alpha \in [0.060, 0.082]$, $\log_{10} T \in [-0.17, -0.04]$: 35,759 of 65,536 cells are chaotic, and periodic windows of periods 2–32 are embedded among them. Rendered as in the shrimp literature [37], periodic regions show dark curves of maximal contraction inside the windows.

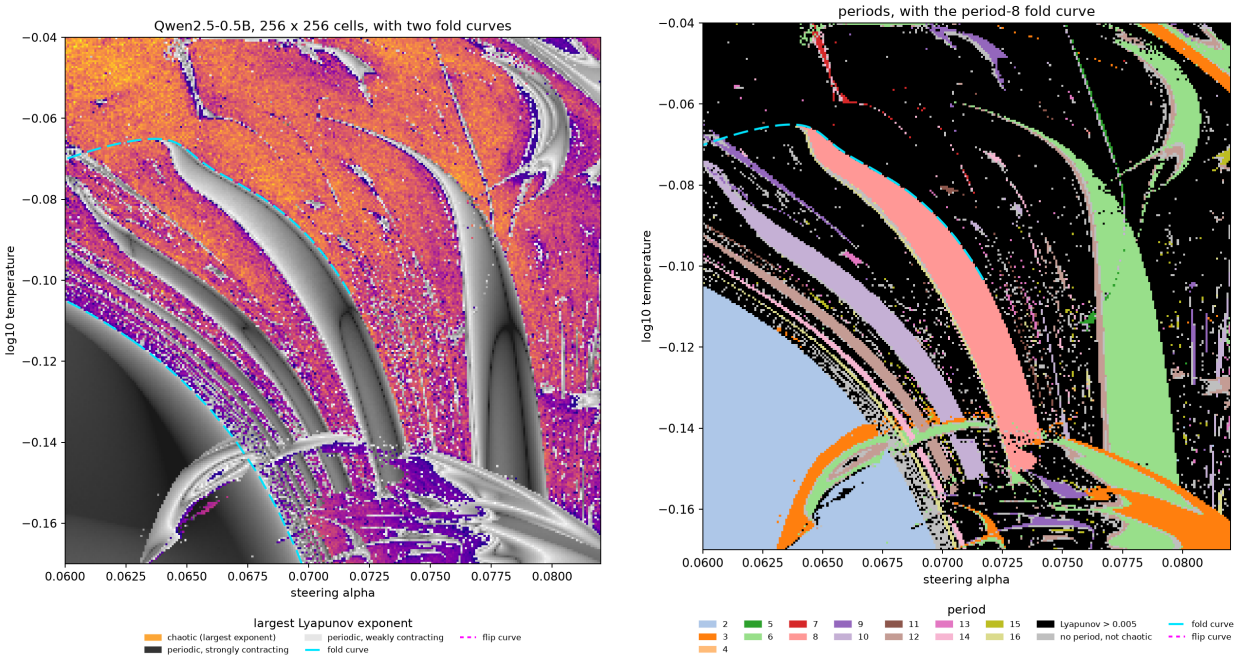


Figure 1: Soft-token map of Qwen2.5-0.5B, 256^2 float64 cells. Left: chaotic cells coloured by the largest Lyapunov exponent, periodic cells in grey by how strongly they contract (light grey: close to losing stability); cyan dashed curves: float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods with the period-8 fold curve (Section 7).

Figure 2 resolves the stripes' lower ends at 512×512 cells (79,278 chaotic). Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 8). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$. Where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

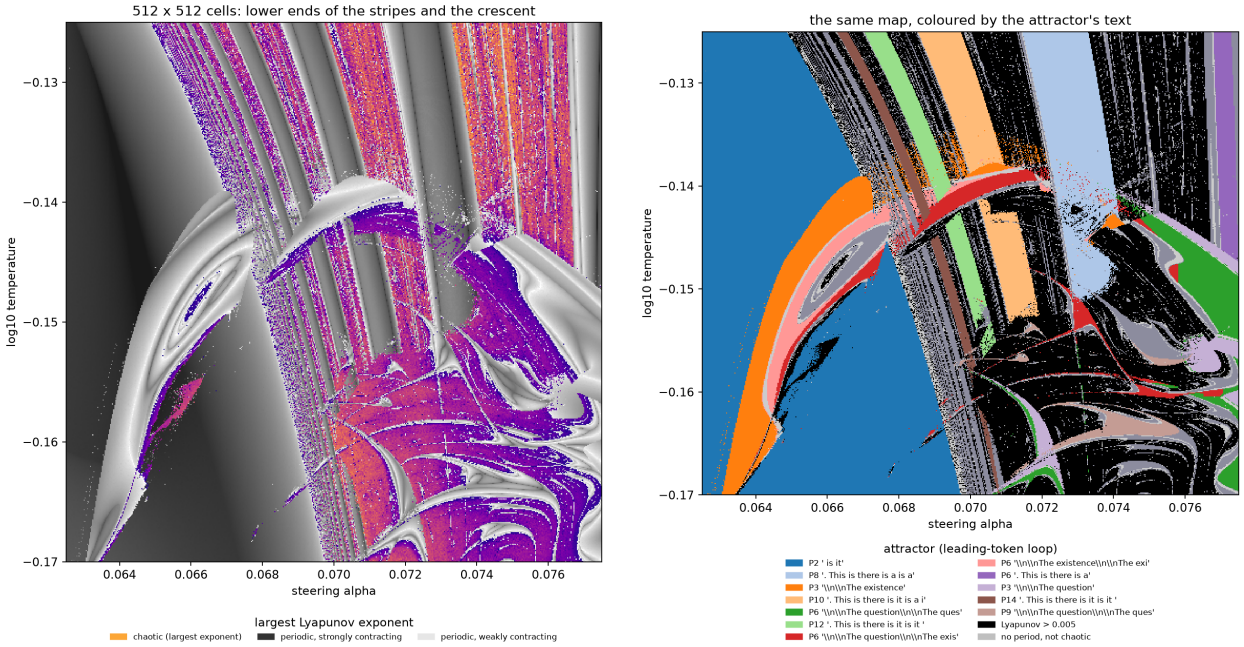


Figure 2: 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$, rendered as Figure 1 (left) and coloured by the attractor's leading-token loop (right; black: chaotic, grey: no period up to 32).

Zooms of 256^2 cells (Figure 3) show what the model does in each window. Every window is a repeating piece of text. The period-8 band repeats ". This is there is a is a", the neighbouring bands add "is it" laps (periods 10 and 12), and along the band's flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the "is X" chain with a paragraph break and "The" (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

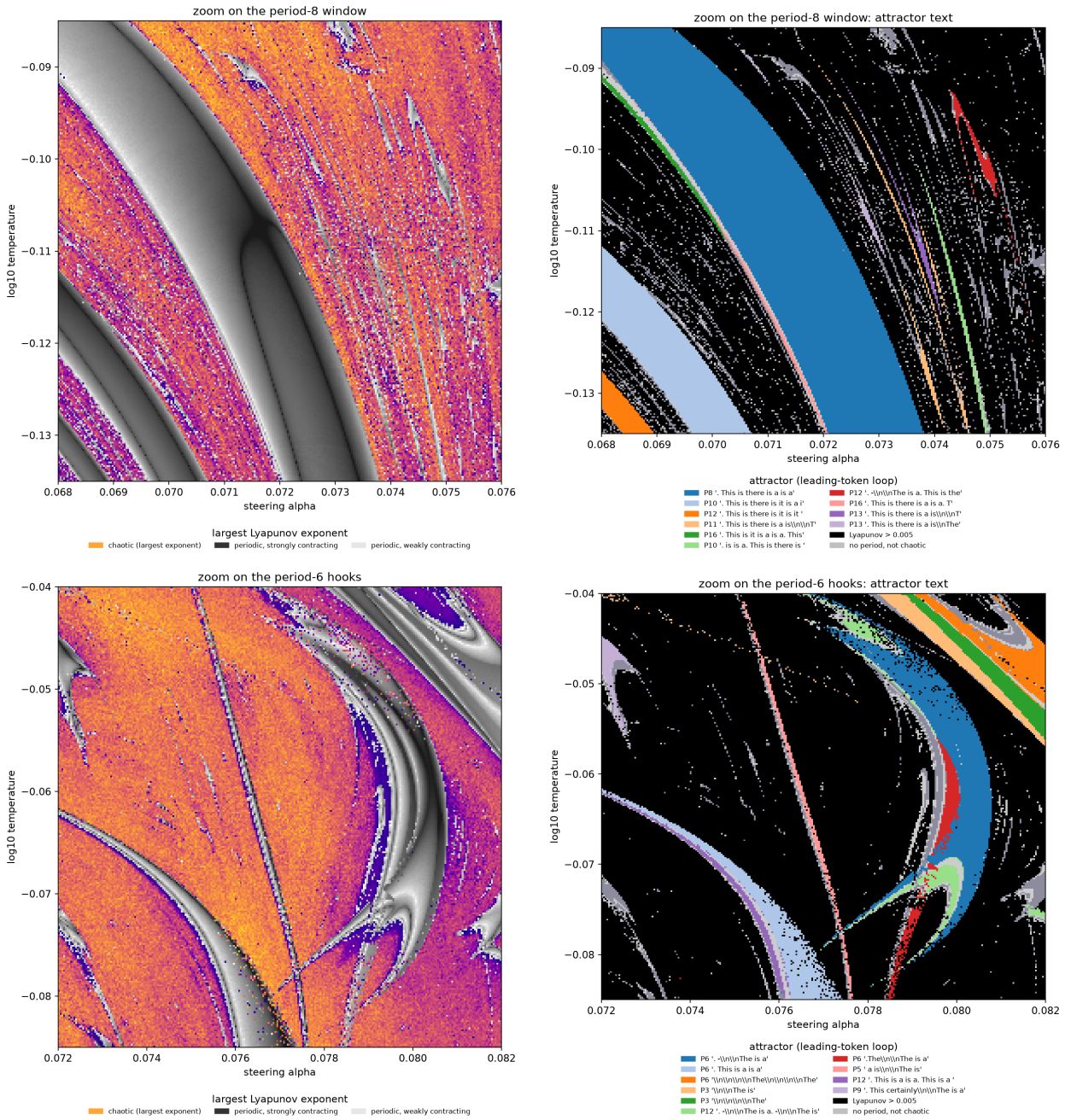


Figure 3: Zooms of 256^2 float64 cells. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analysed in Section 7.

5 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64, read as text (Table 1): ". This" followed by k pairs "is X", period $2k + 2$, each stripe adding one pair. The period-2 region is the 2-cycle "is it". At the stripe centres the multipliers are effectively one-dimensional: the leading one is 0.003–0.09 and the next falls from

3×10^{-6} (period 6) to 2×10^{-14} (period 20). This compression is a property of these orbits, not of the map: the 2-cycle's multipliers are -0.271 and -0.058 (the second modulus is about 0.037 near its fold); at the period-3 flip of Section 8 the next modulus is 0.088; and one period-6 orbit in Section 7 has a complex leading pair.

period	orbit (leading tokens)	μ_1	$\lvert\mu_2\rvert$
6	. This is a is a	$+0.0052$	3×10^{-6}
8	. This is there is a is a	$+0.0110$	3×10^{-7}
10	. This is there is it is a is a	$+0.0025$	2×10^{-7}
12	. This is there (is it)^2 is a is a	$+0.0163$	4×10^{-8}
14	. This (is it)^4 is a is a	$+0.0369$	2×10^{-9}
16	. This (is it)^5 is a is a	$+0.0716$	8×10^{-11}
18	. This (is it)^6 is a is a	$+0.0792$	1×10^{-12}
20	. This (is it)^7 is a is a	$+0.0857$	2×10^{-14}

Table 1: Periodic orbits of the stripe family at the stripe centres along $\log_{10} T = -0.115$ ($\alpha = 0.077872$, 0.071674, 0.067766, 0.065550, 0.064443, 0.063857, 0.063534, 0.063335), computed with the current all-float64 map (norms and rotary tables patched): Newton residuals 6×10^{-15} to 8×10^{-13}, leading multiplier μ_1 and next modulus $|\mu_2|$ from Rayleigh–Ritz values. Each step in period adds one "is X" pair to the text. The 2-cycle "is it" itself has multipliers -0.2706 and -0.0585 at $(\log_{10} T, \alpha) = (-0.17263, 0.061579)$. The period-2 and period-8 orbits were also reproduced by an independent implementation (Section 14).

6 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037). The normal form of G has $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. Extrapolating the stable-branch multipliers, $(1 - \mu)^2/(4(\alpha_c - \alpha)) \to ab$, gives 123.1 independently with a quadratic fit (a linear fit gives 120.6). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 2 unchanged.

Controlled passages. Starting at $u_{\text{in}} = -U$ or 0 ($U = 0.1/|b|$) and counting G-iterates until $u > U$, the measured counts match the iterated normal form within one iterate at four distances (Table 2, Figure 4); a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 2: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

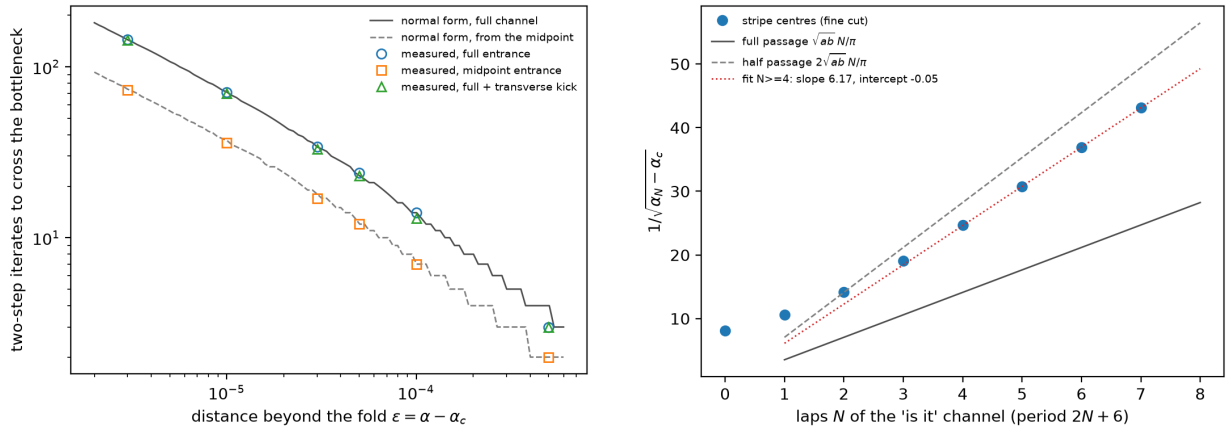


Figure 4: Left: passage times against the normal-form predictions. Right: window law; stripe centres $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($u \approx -3\times10^{-4}$ to -7×10^{-4}). From each measured entry point the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \le 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow the N^{-2} law of a saddle-node ghost with about a third of the full-passage prefactor: stripe centres give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 5) show the near-tangency. Numerically estimated stripe edges along $\log_{10} T = -0.115$ (Table 3) sharpen this. Each window is bounded by a fold, where its orbit is born, and by a flip. For periods 6–12 the fold faces away from the accumulation point; for periods 16–20 the sides are exchanged; the period-14 window is bounded by flips on both sides. The fold edges of periods 16–20 follow $\alpha_N - \alpha_c = C/(N+\delta)^2$ with $C = 0.0266$ and $\delta = 0.15$. Their slope, $1/\sqrt{\alpha_N - \alpha_c}$ rising by 6.13 per lap, is 1.74 times the full-passage slope $\sqrt{ab}/\pi = 3.53$ of the normal form: an effective passage of 0.58 of a full one, as the measured natural reinjection gives independently.

period	edge toward larger α	edge toward smaller α
6	fold 0.0790337	flip 0.0766947
8	fold 0.0726585	flip 0.0706525
10	fold 0.0684572	flip 0.0670594
12	fold 0.0658982	flip 0.0651944
14	flip 0.0645850	flip 0.0642958
16	flip 0.0639284	fold 0.0638001
18	flip 0.0635683	fold 0.0635034
20	flip 0.0633535	fold 0.0633183

Table 3: Edges of the stripe windows along $\log_{10} T = -0.115$ ($\alpha_c = 0.0627983$), from the window centres of Table 1. Flips: critical multiplier a real -1 to 3×10^{-10}. Folds: the multiplier nearest $+1$ extrapolated to 1 through $(1-\mu)^2$, which is linear near a fold; last converged multiplier 0.991–0.998.

[**Planned:** prediction of the window positions from the measured reinjection distribution.]

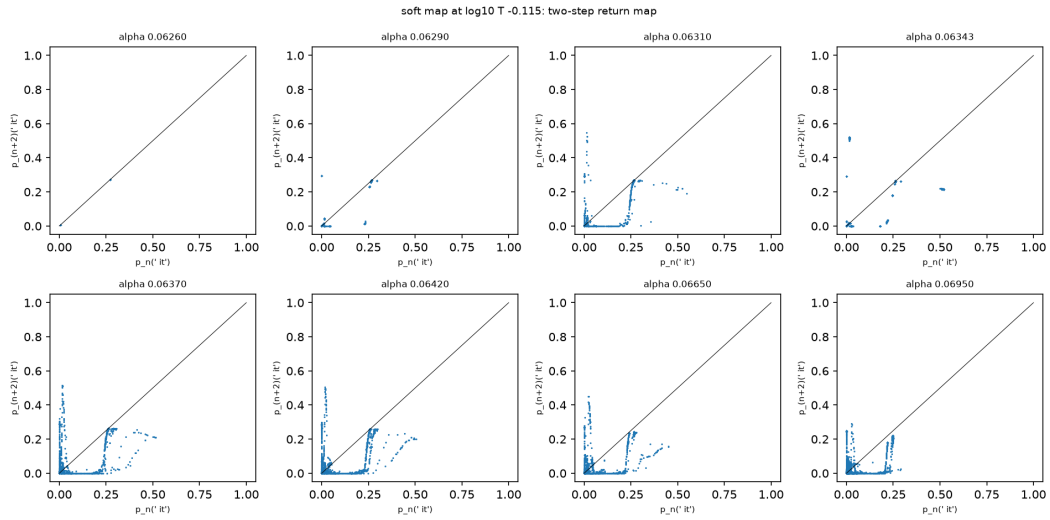


Figure 5: Two-step return maps of the probability of "it" near the 2-cycle's fold, at increasing distance from it.

A prospective test. An independent implementation located two new folds and computed their normal-form coefficients by nested automatic differentiation. Before any test trajectory was run, all 24 passage predictions were pushed in a frozen design. Target A is the 2-cycle's fold at a new temperature: $\log_{10} T = -0.140$, $\alpha_c = 0.0670549$, $a = -7.055$, $b = -24.927$. Target B is the fold of the period-8 orbit of Table 1, continued in temperature at $\alpha = 0.0717$: $\log_{10} T_c = -0.1033462$, $G = F^8$, $a = -11.900$, $b = -152.05$ per unit $\log_{10} T$. Of the 24 forecasts (four distances and three entrances per target), 23 fell within one iterate of the measured passages (Table 4). The pre-specified criterion that every row pass failed by one row: target A, midpoint entrance, $\varepsilon = 3 \times 10^{-6}$, predicted 61 and measured 59. All full-channel passages and all passages after a transverse kick were within one iterate.

ε	target A ($G = F^2$): measured / predicted	target B ($G = F^8$): measured / predicted
3×10^{-6}	118/118, **59/61**, 117/118	26/26, 13/14, 25/26
1×10^{-5}	56/56, 28/29, 56/56	10/10, 5/5, 10/10
3×10^{-5}	26/26, 13/14, 26/26	4/4, 2/2, 4/4
1×10^{-4}	10/10, 5/6, 10/10	2/2, 1/1, 2/2

Table 4: Prospective passage test at two folds that did not shape the mechanism. Each cell lists the full-channel, midpoint and full-plus-transverse entrances as measured / predicted G-iterates. Predictions come from the iterated quadratic normal form, with coefficients from an independent implementation, frozen before the test runs. Bold: the one row outside the one-iterate criterion. The shortest period-8 rows give only weak relative accuracy.

7 Bifurcation curves and cusps

Period-8 stripe. The fold curve of the period-8 orbit, continued for 80 points with a float64 test function, keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge, and caps its top near $\log_{10} T = -0.065$ (Figure 1). At $\alpha = 0.0717$ the stripe lies between a flip at $\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its

multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$; the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. [**Planned:** classify the two reversals on the flip curve (on a flip curve they are not fold cusps).]

The accumulation boundary. The fold curve of the 2-cycle "is it", continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 1 and 2. This is the boundary on which the stripe fan accumulates, as the mechanism of Section 6 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11): a codimension-two point, [**Planned:** to be located and typed].

A period-6 window with the anatomy of a shrimp candidate. The hooks of Figure 3 (bottom) belong to the period-6 loop ". -\n\nThe is a". Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: at the body the multipliers are -0.061 and $+0.048$, lower down -0.564, and on the leg -0.645. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, ".The\n\nThe is a", whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot. Fold curves bounding a window and meeting in cusps, with flip curves inside, is the codimension-two organization of a shrimp [45]. The continued curves (Figure 6) bound the body by a single fold arc on its outer side and a flip arc on its inner side, with the leg between them where they converge. The outer fold arc extends far to the left as a thin arm and ends in a cusp at $(\alpha, \log_{10} T) = (0.06911, -0.03975)$. The fold curve through the close fold pair has a second cusp inside the body, at $(0.08010, -0.05835)$, from which two fold branches descend. At both, the normal-form coefficient changes sign while a stays away from zero (Table 5). The signs of a and b both flip with the orientation of the critical eigenvector, so the test uses $b \operatorname{sign} a$. The cusp's nondegeneracy condition is a nonzero cubic coefficient c of the normal form $u' = u + a\varepsilon + bu^2 + cu^3$ where b vanishes. We compute it with the second-order centre-manifold term, $c = l^{\mathsf{T}} C(r, r, r)/6 + l^{\mathsf{T}} B(r, h_2)/2$ with $(I - M)h_2 = B(r, r) - 2br$ and $l^{\mathsf{T}} h_2 = 0$ (M the monodromy matrix, B and C its map's second and third derivatives, by central differences of float64 Jacobian-vector products; the code reproduces c to 10^{-6} on a map with a known reduction). At both cusps c is nonzero far beyond its step-size uncertainty and keeps its sign through the zero of b, which crosses zero with nonzero slope along the curve: both are nondegenerate cusps, numerically. Its negative sign means that between the fold branches near each cusp two stable period-6 orbits coexist with a saddle.

	cusp on the outer arc	cusp inside the body
$b \operatorname{sign} a$ at consecutive curve points	$+92.9$, $+42.9$, $+14.7$, $\mathbf{-16.9}$, -52.8	-24.7, -14.3, -2.0, $\mathbf{+12.8}$, $+28.0$, $+45.9$, $+58.9$
$\lvert a \rvert$	1.2–6.9	3.13–3.35
c at the two points around $b = 0$	-1.37×10^4, -1.65×10^4	-2.38×10^5, -2.34×10^5

Table 5: Normal-form test at the two parameter-tangent reversals of the period-6 window's fold curves (bold: the flagged points; critical multiplier 1 to 10^{-7}). c: cubic coefficient (finite-difference steps agree to 0.1 and 0.7 %). Numerically validated, not certified.

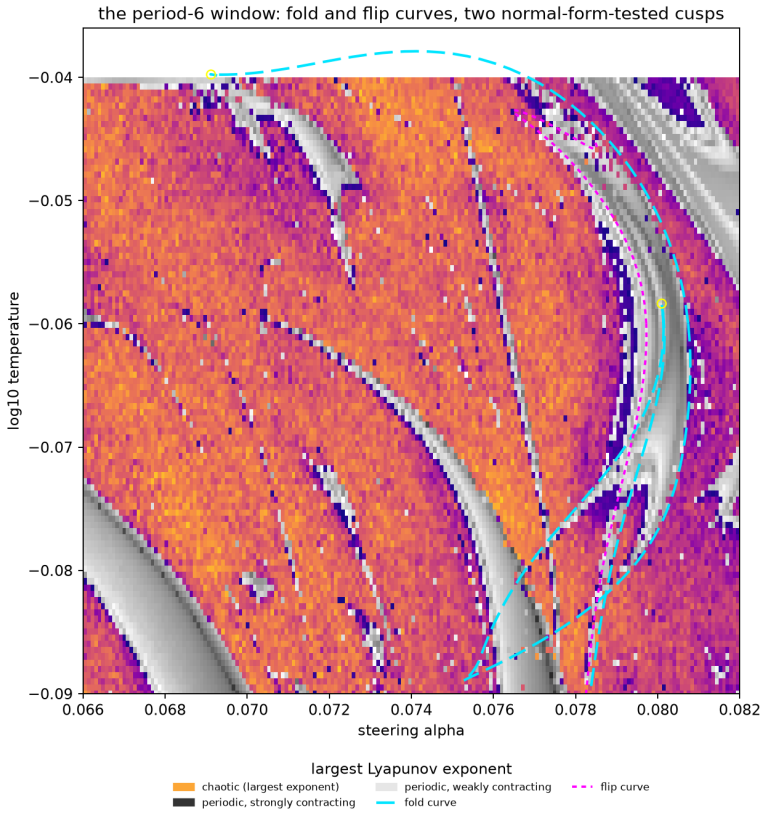


Figure 6: The period-6 window with its float64 fold curves (cyan, long dashes), flip curve (magenta, short dashes) and the two normal-form-tested cusps (yellow circles), over the 256^2 map of Figure 1 (whose data end at $\log_{10} T = -0.04$). [**Planned:** the lower ends of the fold curves, which approach each other near $(0.0753, -0.089)$.]

A period-4 window with two cusps. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). In temperature at $\alpha = 0.06484$ the branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. Continuing that fold in the plane, its parameter tangent reverses twice; at both reversals the normal-form coefficient b changes sign while a stays away from zero (Table 6), so both are numerical cusp candidates of the fold curve (tangent reversal and a sign change of the quadratic coefficient; the cubic nondegeneracy coefficient is not computed), at $(\alpha, \log_{10} T) = (0.06296, 0.05996)$ and $(0.06557, 0.07445)$. Between the two fold branches two period-4 orbits coexist (Figure 7).

	cusp 1	cusp 2
b at five curve points	$+28.1,\ +18.7,\ +8.4,\ \mathbf{-1.1},\ -12.8$	$+51.8,\ +29.7,\ +9.3,\ \mathbf{-9.0},\ -23.4$
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}
c around $b = 0$	$-1.06,\ -1.01$ $(\times 10^4)$	$+1.39,\ +1.36$ $(\times 10^5)$

Table 6: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). c: cubic coefficient, nonzero at both (steps agree to 10^{-4}), so both are nondegenerate cusps numerically; at the second, $c > 0$: the orbit between the two others is the stable one.

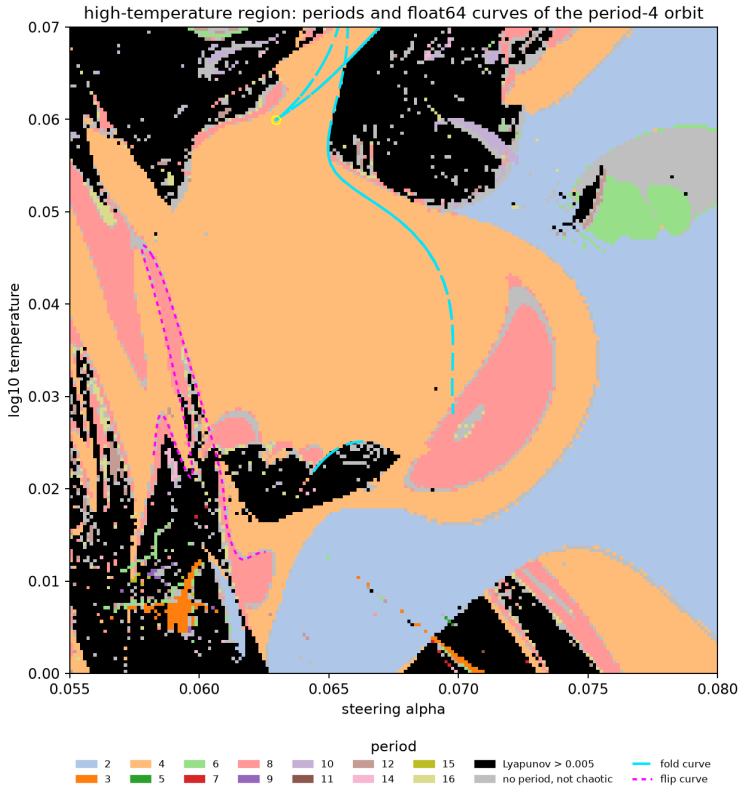


Figure 7: Period map of the high-temperature region (192^2 float64 cells) with float64 fold curves (cyan, long dashes) and flip curves (magenta, short dashes) of the period-4 orbit; the yellow circle is the first cusp (the second lies just above the window). The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where two period-4 orbits coexist. In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core; along the cut of Section 8 the period-8 orbit merges back into period 4 (a doubling bubble), so this band is not shown to be a second cascade. [**Planned:** remaining fold curves (C, D); whether the region is one shrimp-organized component.]

[**Planned:** Codimension-two points where a second multiplier crosses the unit circle along the lower fold curve (candidate fold-flip points): locate, type and continue.]

8 A period-doubling cascade to chaos

The crescent of Figure 2 is a closed region where the period-3 loop "\n\nThe existence" doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4, period 24 in one column, and then a chaotic core. The doubled loops repeat the same leading tokens, twice and four times, with alternating probabilities. Locating the flips in float64 (Section 3) along this cut, in the steering coordinate α, gives Table 8. The first two ratios, $\delta_2 = 2.844$ and $\delta_3 = 2.859$ (indexed as in Table 8), are far from Feigenbaum's 4.6692, although from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, 2×10^{-6}, 2×10^{-12}). The next, $\delta_4 = 3.781$, is markedly larger. The flip parameters agree across three devices (CPU, a consumer GPU and an H100, all float64) to 10^{-16}–3×10^{-15}. Continuing a just-born doubled orbit past period 48 failed: Newton slides onto the half-period orbit, which is also periodic with the doubled period. We therefore also located the superstable parameters

s_p, where the critical multiplier vanishes in the middle of each band (Table 7). There the orbit attracts, so it can be reached by forward iteration and cannot slide onto its half-period parent. Through period 384 the superstable ratios are 3.19, 4.02, 4.478 and 4.625. The last two come from parameters resolved to 10^{-14} and differ from Feigenbaum's value by 0.19 and 0.044; from 4.02 on, the distance shrinks by factors of 3.4 and 4.3. The flip ratios 2.84, 2.86, 3.78 rise in step (Figure 8). The accumulation point lies near $\alpha_\infty \approx 0.0661250$. ($s_{12}$ and s_{24} are known only to within brackets of 1.2×10^{-5} and 7×10^{-7}, because the other computed Ritz values of those orbits blur the root, so the ratios 3.19 and 4.02 have worst-case ranges $[2.90, 3.34]$ and $[4.02, 4.10]$; s_6 is less precise still and s_3 was not bracketed.)

period p	superstable parameter s_p	bracket width	ratio $\frac{s_p - s_{p/2}}{s_{2p} - s_p}$
12	0.0659717	1.2×10^{-5}	
24	0.06608014	6.9×10^{-7}	3.19
48	0.066114132784466	3×10^{-15}	4.02
96	0.066122589892689	7×10^{-15}	4.4779
192	0.066124478510077	6×10^{-15}	4.6251
384	0.066124886853811	5×10^{-15}	

Table 7: Superstable parameters of the crescent's doubling sequence along $\log_{10} T = -0.14798$: the critical multiplier of the period-p orbit changes sign across the bracket, with Newton residuals below 3×10^{-13} and the other computed Ritz moduli below 10^{-20} from period 96 on. Brackets for periods 12 and 24 are the closest points where the critical multiplier is cleanly identified.

One candidate explanation is a crossover. Softmax at temperature T is a smoothed argmax, so the reduced one-dimensional map may have a nearly corner-like extremum, rounded only on a scale set by T. Early doublings would then see the corner, where ratios tend toward 2, and later doublings the smooth quadratic core, where the ratio is 4.669. This predicts ratios rising toward 4.669 at higher periods, and smaller early ratios at lower temperature.

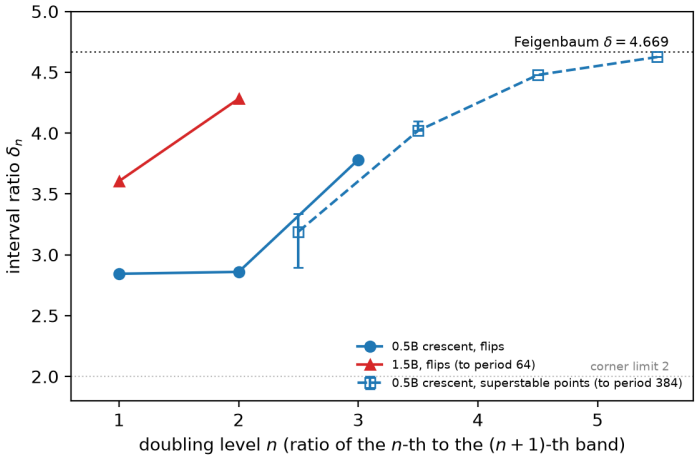


Figure 8: Finite-order interval ratios of the doubling sequences: the 0.5B crescent, from flips and from superstable points (which reach period 384), and Qwen2.5-1.5B from flips. A superstable ratio spans half-bands, so it is plotted half a level after the flip ratio of its first band. Bars: worst-case ranges from the brackets of s_{12} and s_{24}. Each sequence rises toward Feigenbaum's δ from below; the last superstable ratio is within 1 % of it. Under the crossover hypothesis, early levels see the corner of a smoothed argmax (ratios toward 2) and later levels the smooth quadratic core.

[**Planned:** ratios at other temperatures.] Finite-order ratios also depend on the parameter coordinate, so only the trend of later ratios can test universality [31, 32, 47].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	2.859
4	$24 \to 48$	0.0661071412523	2.3×10^{-12}	3.781
5	$48 \to 96$	0.0661209563174	$< 10^{-13}$	

Table 8: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. For the five displayed H100 flips the critical Ritz value is real and differs from -1 by at most 4×10^{-10}, the Newton residuals are below 4×10^{-13}, and the other computed Ritz values lie inside the unit circle. Feigenbaum's constant is $\delta = 4.6692\ldots$ [47]; for the logistic map the first ratios are 4.751, 4.656, 4.668. The first four parameters agree between the completed CPU and H100 computations to 4×10^{-15}. Superstable parameters (Table 7) extend the ratio sequence through period 384.

In the high-temperature region (Figure 7), along $\log_{10} T = 0.0319$, the period-2 loop "\n\nThe" flips at $\alpha = 0.0753534$ and its doubled loop at 0.0736921. These orbits are far from one-dimensional (next moduli 0.38 and 0.28). Toward smaller α the period-8 orbit does not double again: near $\alpha \approx 0.0705$ it merges back into the period-4 orbit. Along this cut it is therefore a period-doubling bubble, bounded by two flips of the period-4 orbit, not the start of a cascade to chaos.

9 Bridge to sampled decoding

Replacing the expected embedding by the mean of m sampled token embeddings (Figure 9): with the exact map, laminar phases near the fold last 120 and 63 steps at $\varepsilon = 10^{-5}$ and 3×10^{-5}; with $m = 64$ they last about 10.6 steps independently of ε (57% of steps in the channel); with $m = 8$, about 7.7 steps (38%); with single samples ($m = 1$, sampled decoding of the same truncated system) essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism belongs to the expected-embedding map; sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

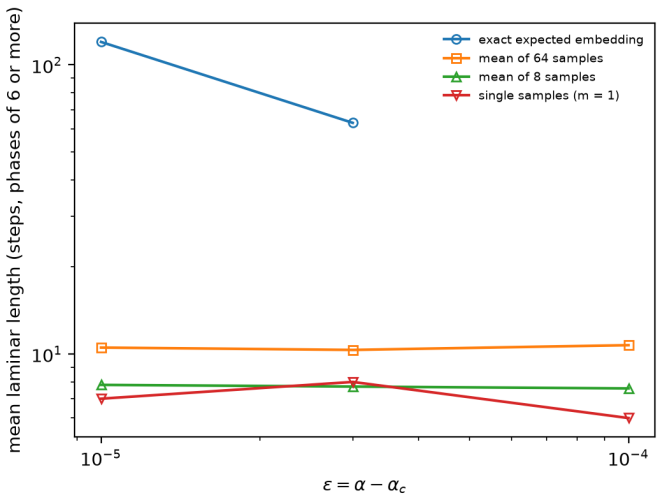


Figure 9: Mean laminar length against ε for the exact map and for m sampled embeddings.

10 Replication and scale

For Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the same range again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 13) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic windows inside it. The large periodic regions are question loops: "What?\n" (period 2), "\nWhat is a" (period 4) and "What continuum?\n" (period 3). This window has no period-adding fan like the 0.5B model's. A cut of 2048 cells at $\alpha = 0.100$ (Figure 10) shows how the band is entered from below. The 2-cycle "What?\n" gives way abruptly to a period-4 loop: the Lyapunov exponent jumps, so these are two distinct attractors, not a flip. The period-4 loop then doubles: float64 flips at $\log_{10} T = -0.0031269$ (period $4 \to 8$, next modulus 0.014), -0.0021666 (period $8 \to 16$, next modulus 8×10^{-5}) and -0.0019002 (period $16 \to 32$, next modulus 5×10^{-9}) and -0.0018380 (period $32 \to 64$), where the exponent rises continuously to zero; chaos follows. The interval ratios are 3.60 and 4.28. They rise toward 4.669 as in the 0.5B crescent, and from higher values, at higher temperature ($T \approx 0.99$ against 0.71), as the crossover hypothesis of Section 8 would have it. With different models and levels, the comparison is only suggestive.

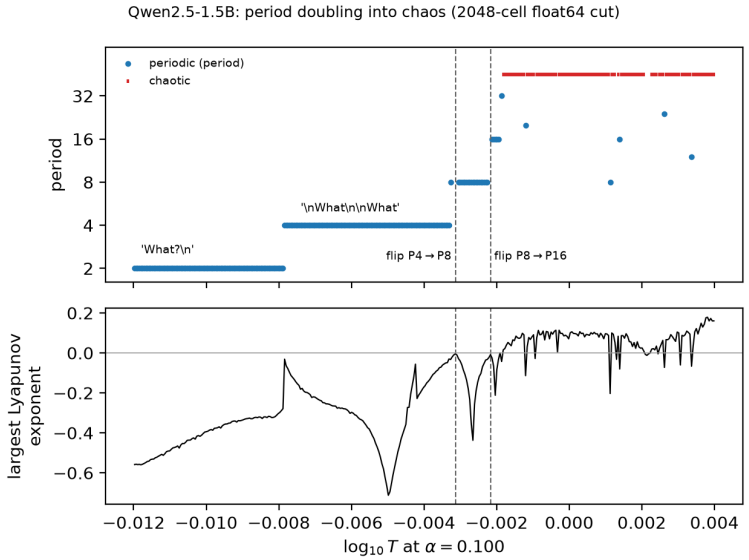


Figure 10: Qwen2.5-1.5B along $\log_{10} T$ at $\alpha = 0.100$: period of the attractor reached from the fixed start (top) and largest Lyapunov exponent (bottom). Dashed lines: float64 flip parameters. The doubled loops alternate the number of line breaks between the "What"s.

(A second steering direction, hostility, reproduces the fold-and-adding mechanism; Section 12.)

For Qwen2.5-7B (steering at layer 14 of 28, concept vectors built as for the smaller models), coarse 64^2 float64 maps show the same kind of organization (Figure 11). Over the philosophy range used for 0.5B and 1.5B, 8 % of cells have a positive largest exponent, and periods 2, 4, 3 and 11 dominate the periodic cells. Along the hostility direction ($\alpha \in [0, 0.12]$, $\log_{10} T \in [-0.3, 0.3]$), 24 % of cells are chaotic, with bands of period-3, -5 and -6 windows in a chaotic sea below $\log_{10} T \approx 0.1$ and high-entropy chaos above. The windows hold genuine periodic orbits: in float64, Newton's method converges at the most interior period-6 cell to the loop ", shut the end\n\nOh" at $(\log_{10} T, \alpha) = (-0.0143, 0.0476)$ (residual 2×10^{-14}; spectral radius 1.6×10^{-7}, nearly superstable) and at the most interior period-3 cell to a period-3 loop at $(0.2524, 0.1143)$ (residual 3×10^{-15}; leading multiplier -0.12). Continued for about an hour each, over ± 0.001 in α and -0.016 in $\log_{10} T$ (one float64 continuation step takes about two minutes at 7B), both orbits stay strongly stable without a crossing. [**In progress:** continuation across the period-6 window's edge, where the map switches to the period-3 loop "you better than".]

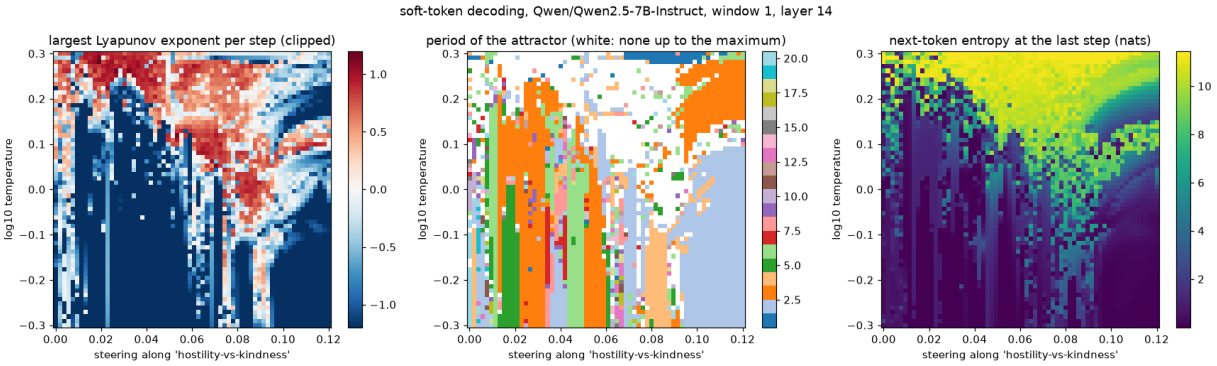


Figure 11: Qwen2.5-7B soft-token map steered along the hostility direction (layer 14), 64^2 float64 cells: largest Lyapunov exponent, period of the attractor reached from the fixed start, and next-token entropy at the last step.

A second model family. OLMo-2-1124-7B-Instruct [49] comes from another laboratory, with other data, training and code. Steered at layer 16 of 32, with concept vectors built exactly as for Qwen, it shows the same kind of organization (Figure 12). In 64^2 float64 maps, 6 % of cells are chaotic along the philosophy direction and 13 % along the hostility direction, in a band near $\log_{10} T = 0.03$–0.17 that bends to lower temperatures under strong hostility steering ($\alpha > 0.07$). Below the band the attractor is a fixed point or the 2-cycle "!Hello"; inside it lie periodic windows of periods 4 to 31. Along the hostility direction the largest windows, of periods 8 to 12, are loops of k laps of the 2-cycle "!<|endoftext|>" with a short excursion ("there is an"), so each lap adds two to the period, as each lap of "!Yes" does in the hostile cascade of Qwen2.5-0.5B (Section 12). Whether a fold of that 2-cycle organizes them is not tested; these are coarse pixels, and no OLMo orbit has been continued.

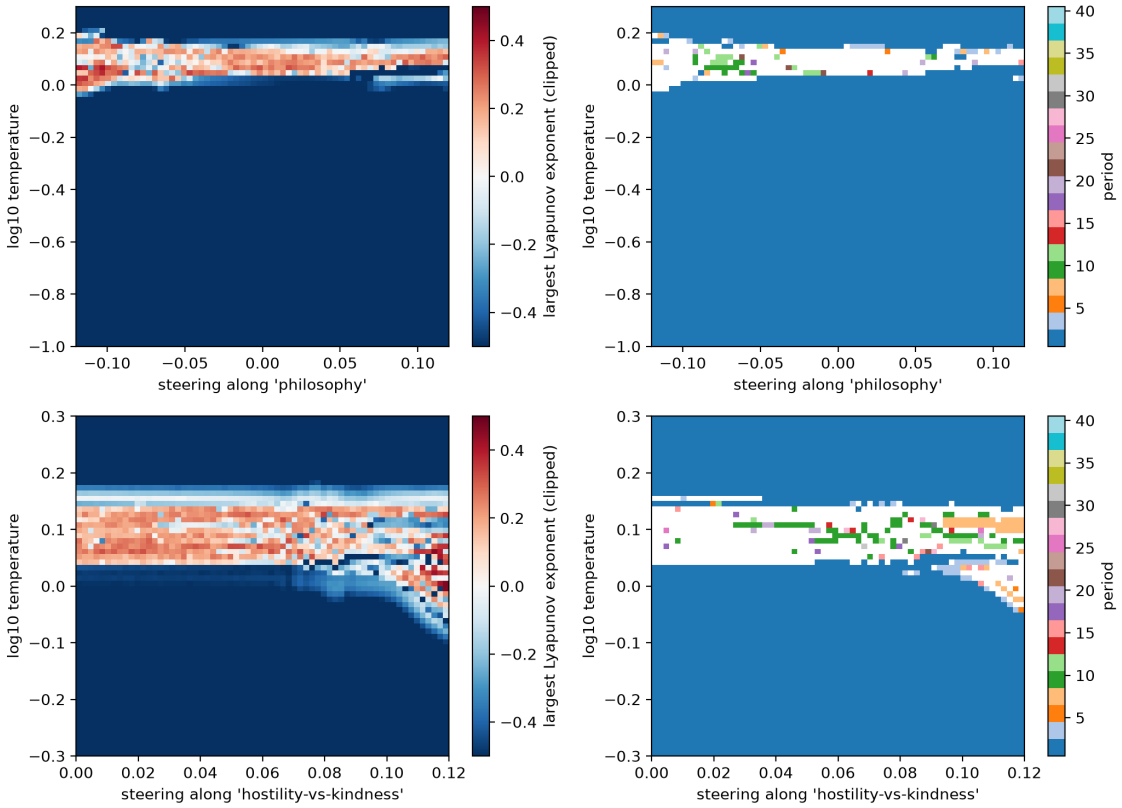


Figure 12: OLMo-2-1124-7B-Instruct soft-token maps, 64^2 float64 cells each, steered at layer 16 along the philosophy direction (top) and the hostility direction (bottom): largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to the maximum).

[**Planned:** scaling of ab and of the window law with model size.]

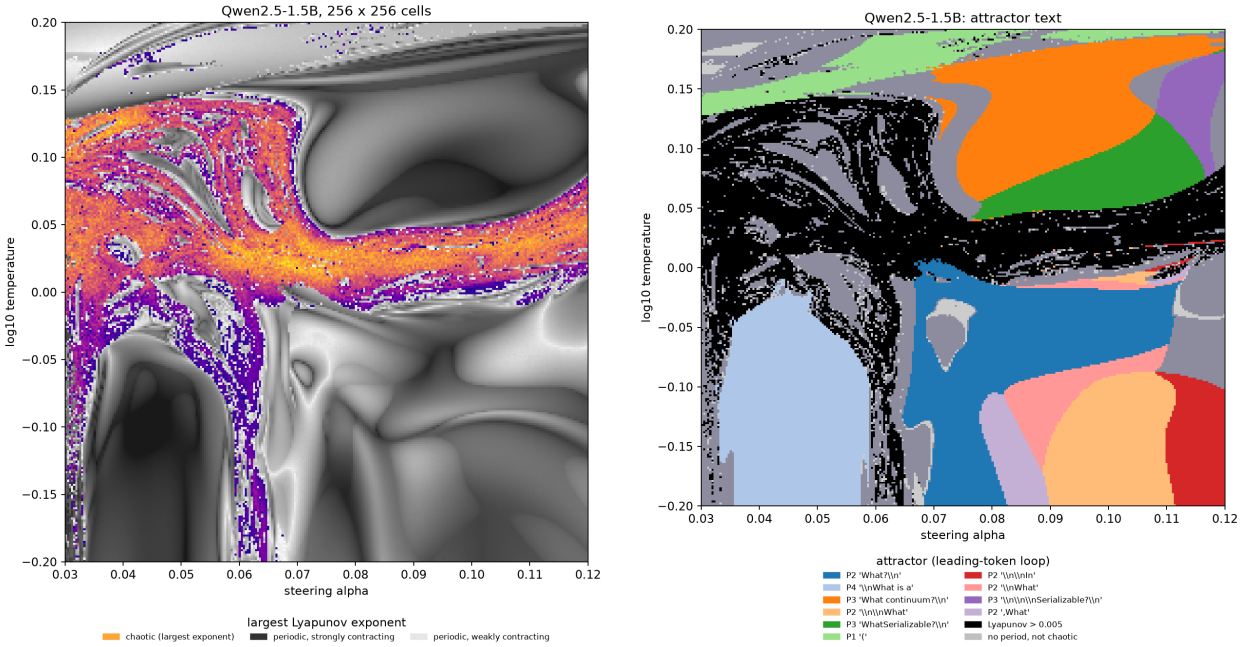


Figure 13: Qwen2.5-1.5B soft-token map, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$, rendered as Figure 2. Several period-2 label regions touch without a gap; as in Section 3, such boundaries may be label switches of one orbit.

11 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [23]. For a GSM8K-style prompt the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$ this predicts a Neimark–Sacker bifurcation at $\lambda_c = 2(1 - \mathrm{Re}\,\mu)/|1-\mu|^2 = 1.0777$. The latent state decays below it and settles on a closed curve above it; a fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$ (supercritical). Causal attention makes the loop block lower-triangular in token positions; the first position to oscillate is the "." of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [13]; the predicted crossing, its amplitude law and the rotation number are what this adds.

The spectrum predicts the relaxation response. With the full instruction template, the GSM8K prompts' fixed points look different: the leading multipliers surround the origin, a real 0.914, complex pairs of modulus 0.85–0.89 at many angles and negative real values near -0.8, with no near-critical pair. Because the prefill Jacobian is block lower-triangular in positions, its spectrum is the union of per-position blocks, and template positions contribute identical eigenvalues to every prompt. Such a spectrum makes three predictions before any task is scored. Within the computed spectral estimates of these prompts, no fixed relaxation can speed convergence much: with leading

multipliers spread around the origin at many angles, any λ that shrinks one group enlarges another, and the best $\lambda^* = 1.03$–1.06 lowers the rate from 0.914 to only 0.908–0.911. (A finite computed spectrum does not show that no other polynomial scheme could do better; Anderson acceleration, tried below, did not.) Under-relaxation slows it ($\lambda = 0.55$: predicted 0.953, measured 0.951 on one prompt). Over-relaxation destabilizes the negative modes first, at $\lambda_c = 2\,\mathrm{Re}(1-\mu)/|1-\mu|^2 = 1.074$–$1.126$ (five prompts), a flip or a Neimark–Sacker bifurcation close to -1. The approach to it is visible: on the prompt with $\lambda_c = 1.074$, the measured rate at $\lambda = 1.05$ is 0.955 against a predicted 0.956 (native 0.864). Elsewhere the spectral radius bounds the rate measured over the accessible window from above, by up to 0.05, because the slowest modes carry little amplitude there. Independently, Wang et al. [50] report a Huginn GSM8K drop from 19.41 % to 8.34 % at a constant step of 1.2; at 0.8 GSM8K is 19.79 %, while their four-task mean falls slightly. Their trajectory-based scheduling analysis does not compute the prompt-specific spectral thresholds studied here.

Does the task notice? (development data). On ARC-Challenge questions scored by log-likelihood, where one prefill per answer choice is exactly the analysed recurrence, the spectra again predict convergence: rates agree to 0.002–0.006 where the slowest mode dominates, and beyond each question's λ_c (1.075–1.149 for four questions) the residual stops decaying. For 6 of 30 development questions the native fixed point is itself unstable ($\rho(1) = 1.009$–1.273; the fixed point, located with a relaxed transient, has a real multiplier beyond -1 or a complex pair outside the circle), so the native loop oscillates (relative amplitude 2×10^{-2} for one of them), and relaxation to 0.55 or 0.75 restores convergence to 10^{-6}. For the other 24 the native loop is stable up to $\lambda = 1.020$–1.149 (median 1.095). Question by question, an estimated spectral feature, $\hat{\rho}_q(\lambda)$ (the largest $|1 - \lambda + \lambda\mu|$ over the computed Ritz values of the question's gold-answer prefill), ranks the question's residual (Spearman 0.73 at $\lambda = 1$ and 0.75 at 1.10 over the 30 questions; development data). In this historical analysis the residual was the all-choice aggregate entering the 32nd recurrence call, so this is a predictive association across related maps and precisions, not a same-state contraction identity. The frozen test uses matched gold-row token IDs and a per-row residual with an explicit clock. This is consistent with the latent orbits that Geiping et al. [23] observed on some tokens of harder questions, here traced to the fixed point's spectrum. Accuracy, however, tolerates the instability (300 training questions, 32 steps): 0.367 native, 0.387 at $\lambda = 1.10$ and 0.373 at 1.15, where the latent state oscillates with relative size 0.08 and 0.23; only at 1.25 (size 0.68) does it fall, to 0.300. The oscillation does reach the answers: reading every step from 24 to 32 on 100 questions, the margin between the correct choice and its best rival moves by 0.11 nats per step natively, 0.37 at 1.10 and 0.95 at 1.15, in alternating directions, and the chosen answer changes within these nine steps for 12 %, 16 % and 33 % of the questions. The changes go both ways, so the accuracy at any one step stays the same on average. Under-relaxation instead costs 4–5 points at 8–16 steps and catches up by 32 (slower transient), and Anderson acceleration neither speeds the transient nor gains accuracy. A test of the accuracy consequence needs a task in which such step-to-step changes can compound. GSM8K generation is such a task. With eight chain-of-thought examples in the lm-evaluation-harness format, greedy decoding and 32 recurrence steps per token (development problems 0–119, every arm on the same problems and initial states), accuracy is 0.375 natively, 0.425 at $\lambda = 1.05$ and at 1.10, 0.29 at 1.15 (17 problems lost and 7 gained, one-sided McNemar $p = 0.03$) and 0.16 at 1.20 (29 lost, 3 gained, $p = 10^{-6}$). Under-relaxation to 0.8 costs nothing at 32 steps (0.41). The break lies between 1.10 and 1.15, at or just above the spectral thresholds ($\lambda_c = 1.074$–1.126 for the GSM8K prompts, median $\lambda_{\mathrm{up}} = 1.09$ for ARC questions), and generation breaks earlier than log-likelihood scoring, consistent with answer changes that compound over tokens. The collapse at 1.2 is the one Wang et al. [50] reported. A frozen test on fresh problems confirms

this (design and analysis pushed before any test item was read, after two rounds of independent review). On 200 GSM8K test problems, accuracy is 0.330 natively, 0.395 at $\lambda = 1.05$, 0.255 at 1.15 and 0.095 at 1.20. The two pre-stated predictions pass: accuracy declines at 1.20 (-23.5 points; 50 problems lost and 3 gained, exact one-sided McNemar $p = 3 \times 10^{-12}$), and 1.05 loses no more than four points (difference +6.5, one-sided 95 % lower bound +2.5). At 1.15 accuracy falls by 7.5 points ($p = 0.01$, reported descriptively). Development and test agree closely (Figure 14).

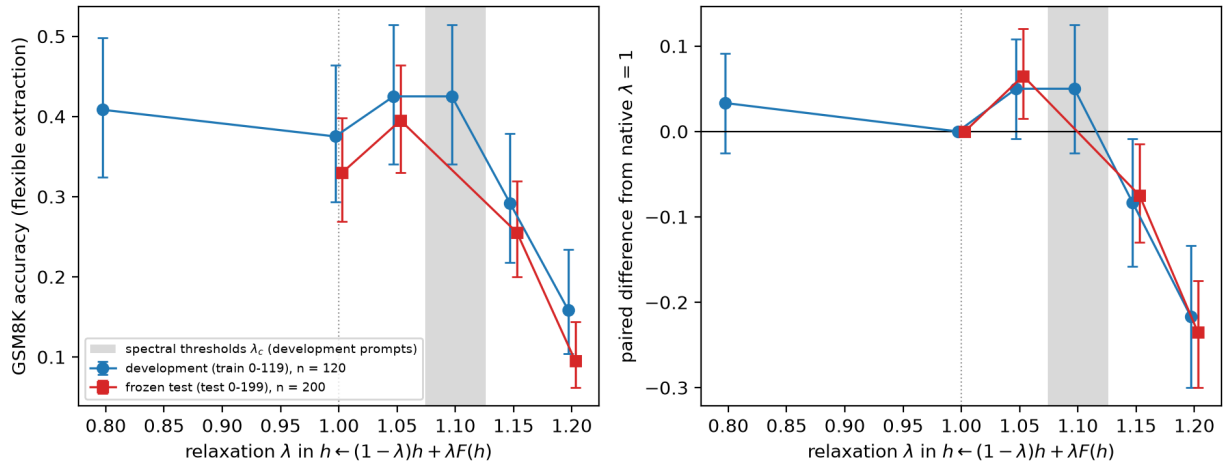


Figure 14: Huginn-0125 chain-of-thought accuracy on GSM8K against the relaxation strength λ (eight examples, greedy decoding, 32 recurrence steps per token): development problems (blue) and the frozen test on held-out problems (red; design and analysis fixed before any test problem was read). Grey: the spectral thresholds $\lambda_c = 1.074$–1.126 computed from the fixed points of five development prompts. Left: accuracy with Wilson 95 % intervals. Right: paired difference from the native loop with paired-bootstrap 95 % intervals.

[**In progress:** the item-level part of the frozen test (spectral feature of each held-out ARC question against its own residual).]

12 Misaligned behaviour as an attractor

The maps show how a smooth change of a control parameter can create, destroy or exchange attractors, and how behaviour near the boundaries becomes intermittent and history-dependent. The same questions can be asked of undesirable behaviour in ordinary sampled dialogue, where attractor-like patterns and persona drift have been observed [9, 51, 52] and input context can push behaviour progressively [53]. Do states such as hostility toward the interlocutor behave like basins of attraction, with capture thresholds, hysteresis and early-warning signals? (Misalignment produced by fine-tuning [54] is a different, weight-level phenomenon.) Appendix A shows the coarse-grained instrument on a benign direction. Self-dialogue of the same model, sampled at $T = 0.7$ and steered along a gratitude direction, falls into a metastable gratitude set. The capture is gradual at $\alpha = 0.01$ (median entry turn 7), fast at 0.02 (median 5) and imposed from the first reply at 0.04, where the set is effectively absorbing within 40 turns.

For hostility we use a valence direction (contrastive hostile minus matched kind statements; its cosine with the gratitude direction is -0.04) and a hostility score from a pinned toxicity classifier (toxic-bert of Detoxify [55]; a reply counts as hostile when its toxic output exceeds 0.5). On the prompt sets the classifier's toxic output is 0.95 for the hostile statements against at most 0.002 for the kind and neutral ones. In a dry run (8 dialogues of 10 turns per dose) no turn was hostile at

$\alpha = 0.02$; at 0.04, 7.5 % of turns were, in bursts and late captures; at 0.08, 87.5 % were, from the first turns. Activation-based methods already anticipate response behaviour [52, 56, 57], committors have been used in language-model decoding [58], and adaptive timing of counter-steering gave no consistent advantage in one persona setting [59]. The primary question is therefore incremental. Near the threshold, does a dynamical estimate add several-turn warning of capture over strong baselines that see the same turns (the classifier's score, its change and history, the turn index and a fixed persona-axis projection)? The dynamical estimates are a Markov-chain hitting probability, a finite-horizon committor [60, 61], and slowing-down indicators [62]. Barrier-crossing rates would follow Kramers' law [63] only under a justified stochastic reduction, and slowing-down proxies are known to depend on the mechanism [64, 65]. A secondary question is whether interventions timed by those estimates prevent capture at lower cost than scheduled or constant counter-steering. Early warning is not expected for noise-induced jumps without a nearby bifurcation, or for transitions faster than the slowing-down; such a monitor would complement classifiers, not replace them.

The hostility direction in the deterministic map. Steering the soft-token map of Section 2 along the same hostility direction gives a plane with chaos (a positive largest exponent in 10 % of a coarse 128×192 float32 scan) and periodic text loops that are hostile: "Yes, that's crap" (period 6), "a lie is" (3), "STUP…" (3), a loop on the Chinese word for insult and its 2-cycle, and others, at α above about 0.035 (the insult loop) and above 0.065 (the rest). Around the period-6 loop the windows form a dome (Figure 17). Toward higher temperature its texts gain one "!Yes" after another (periods $6, 8, \ldots, 20$). In float64 the windows of periods 12 to 20 are born in folds on their low-temperature side and end in flips or $+1$ crossings on the other, like the stripes of Sections 5 and 6; the period-6 window ends in a fold at $\log_{10} T = -0.0991$, and the continuations of the period-8 and period-10 windows failed to converge. The windows accumulate on the fold of the 2-cycle "!Yes" at $\log_{10} T_c = -0.0462465357$ ($\alpha = 0.083$; critical multiplier 0.9999999954, next modulus 0.050). From period 16 on, $1/\sqrt{\log_{10} T_c - \log_{10} T_{\text{fold}}}$ rises by 2.235 per lap. The fold's normal form ($a = -1.168$, $b = 44.53$ with ε in $\log_{10} T$) predicts $\sqrt{|ab|}/\pi = 2.296$ per lap for full passages through the bottleneck, within 3 % and without a fitted parameter. An independent float64 forward and automatic-differentiation implementation reproduces this fold (location to 6×10^{-10} relative, a simple real $+1$ multiplier with next modulus 0.050) and its coefficients (ab to 8 parts per million). In that implementation, controlled passages from fixed entrances at $\log_{10} T$ offsets -10^{-5}, -3×10^{-5} and -10^{-4} took 119/59, 61/31 and 27/13 iterates of F^2 (full channel/from its midpoint), against prospectively frozen quadratic forecasts of 118/61, 61/32 and 27/14. Five of six are within one iterate; the smallest-offset midpoint missed by two, failing the frozen all-six criterion. Finite-endpoint (arctan) forecasts, also frozen, were within 0.923 iterates. These are controlled local transits. At the window distances the finite-channel correction to the asymptotic passage time is large, so the agreement of the window slope with the full-passage coefficient does not by itself show where the natural dynamics reinjects. The same type-I intermittency that organizes the philosophy-steered map thus generates a hostile cascade. A second region, near $(\alpha, \log_{10} T) = (0.06, 0.20)$, shows arms of period-7 and period-8 windows spiralling around the insult loop. In float64 the insult fixed point doubles into a 2-cycle at $\alpha \approx 0.041$ and 0.079, and that 2-cycle has a Neimark–Sacker bifurcation in the spiral region: at $\alpha = 0.062$ its complex pair crosses the unit circle near $\log_{10} T = 0.1656$, turning 0.12 of a revolution per step of the 2-cycle. This does not explain the arms: the two invariant circles born at a 2-cycle's Neimark–Sacker point are exchanged by the map, so orbits locked on them have even periods, near 16 for this rotation, whereas the arms include period 7; and of four float32 arm cells, only one kept its period in float64. Float64 zooms show more structure. In one arm a period-7 domain carries period-14 crescents with period-28 cores along its lower edge, between

period-1 and period-2 regions and a chaotic band. In the cascade the windows are nested crescent arcs inside a dome bounded by the "!Yes" 2-cycle region, and near $(\alpha, \log_{10} T) = (0.087, -0.069)$ the period-12 and period-14 windows bend into hooked, nested arches. These are the map-level features that accompany shrimp (a periodic body with doubling crescents; hooked windows). In float64 the period-7 body is bounded by folds: at $\log_{10} T = 0.1925$ the orbit exists between $\alpha = 0.065966$ and 0.071188, and at $\alpha = 0.066$ it ends in a fold at $\log_{10} T = 0.19263$. Below it the multiplier passes through -1 twice (at 0.1919 and 0.1899), so the period-14 crescent is a doubling bubble, and further down the branch folds twice (0.18797 and 0.18921, three coexisting period-7 orbits in between) before a third flip. All crossings are simple (next modulus below 1.5×10^{-3}). An S-shaped pair of folds is what a cusp of the fold curves leaves on a cut nearby. Continuing these boundaries in the plane in float64 (Figure 16), each of the two fold curves reverses its parameter tangent once. There the oriented quadratic coefficient b of the fold's normal form passes through zero (upper fold: -23.9 to $+244.6$ between adjacent points, with $a \approx 2.0$; lower fold: $+31.2$ to -58.7, with $a \approx 0.55$), while a stays away from zero: two numerical cusp candidates, at $(\alpha, \log_{10} T) \approx (0.0672, 0.1990)$ and $(0.0666, 0.1894)$. The flip curve of the period-14 bubble lies inside the body and closes around the map's period-14 cells. The cubic coefficient is nonzero at both points: $c \approx -1.4 \times 10^6$ to -2.0×10^6 (step agreement 5 %; the smallest steps converge like the square of the step, and extrapolation leaves the sign) and -5.8×10^5 to -5.3×10^5. Both cusps are therefore nondegenerate numerically, and $c < 0$ agrees with the three coexisting period-7 orbits, two of them stable, found between the S-shaped folds. The hostile period-7 body has the same skeleton as the period-6 window of Section 7.

Its interior is a shrimp's too. Float64 bifurcation diagrams through the body (Figure 15; 1400 cells per line) show, at $\log_{10} T = 0.1830$, the period-7 orbit doubling to periods 14 and 28 and entering chaos (largest exponent up to $+0.1$) for $\alpha \approx 0.0659$–0.0673, with windows of periods 28, 35 and 42 inside the chaotic band, then undoubling through 28 and 14 back to 7; the exponent touches zero at every doubling. Newton's method converges at map cells to primitive orbits of periods 14, 28 and 21 (residuals 3×10^{-15}, 2×10^{-14} and 5×10^{-13}; spectral radii 0.08, 4×10^{-4} and 0.006). Periods 21, 35 and 42 are three, five and six times the body's period, the satellite windows expected in the chaotic bands of a period-7 cascade. Continued in temperature, the period-14 orbit flips into period 28 near $\log_{10} T = 0.1841$ and back near 0.1857, the two ends of a doubling bubble that holds period 28 and chaos, and lower down its branch is S-shaped (folds near 0.1824 and 0.18244, three coexisting period-14 orbits between them) before another flip near 0.1819; all these crossings are simple (next modulus below 4×10^{-5}).

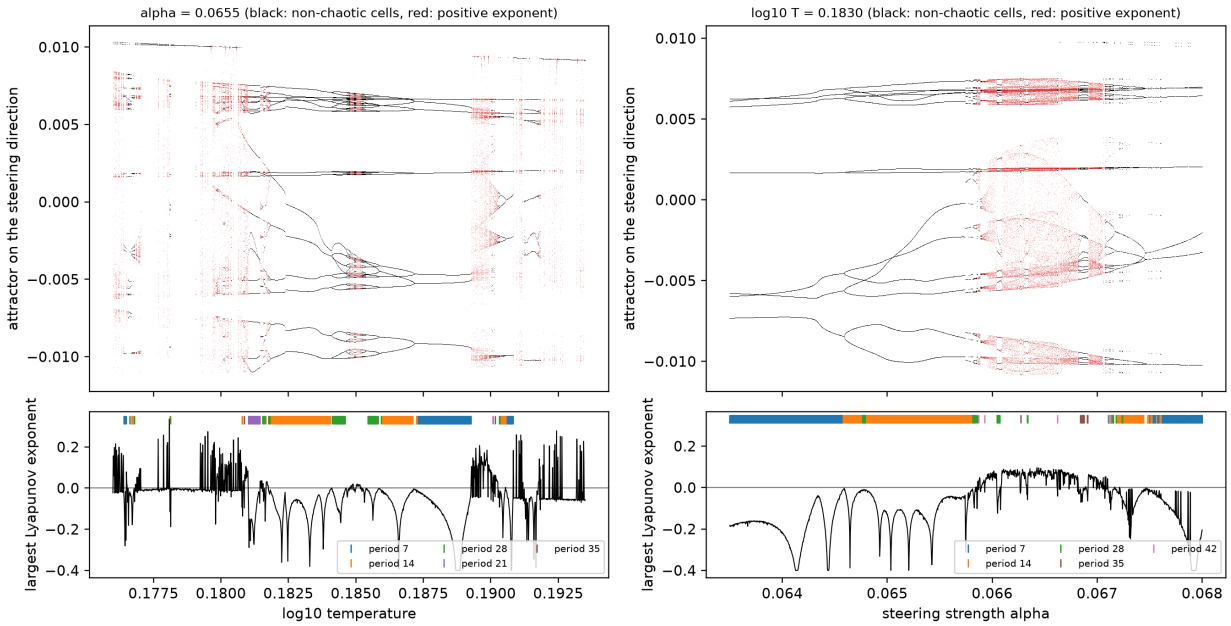


Figure 15: Inside the hostile period-7 body: float64 bifurcation diagrams along temperature at $\alpha = 0.0655$ (left) and along the steering strength at $\log_{10} T = 0.1830$ (right). Top: the last 160 states of the attractor reached from the fixed start, projected on the steering direction (red: cells with a positive largest exponent). Bottom: largest Lyapunov exponent, with the cells' periods marked. On the right, period 7 doubles to 14 and 28, enters chaos with windows of periods 28, 35 and 42, and undoubles back to 7.

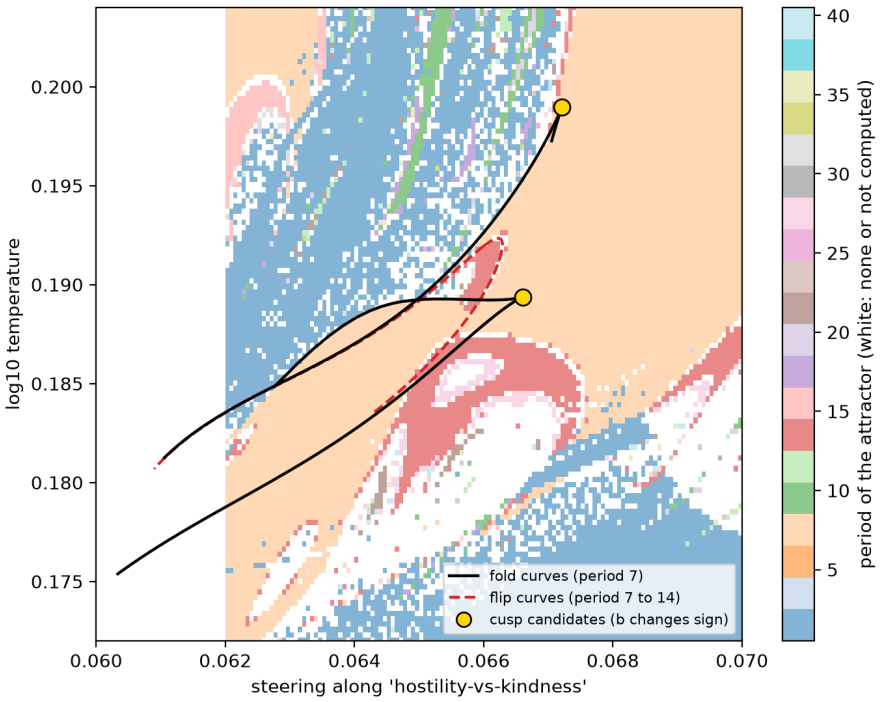


Figure 16: The hostile hub's period-7 body (Qwen2.5-0.5B, hostility steering): attractor periods of a float64 zoom (partial: the top rows were not computed), the float64 fold curves of the period-7 orbit (black) and the flip curves into period 14 (red, dashed), with the two cusp candidates at which the fold's quadratic normal-form coefficient changes sign (gold).

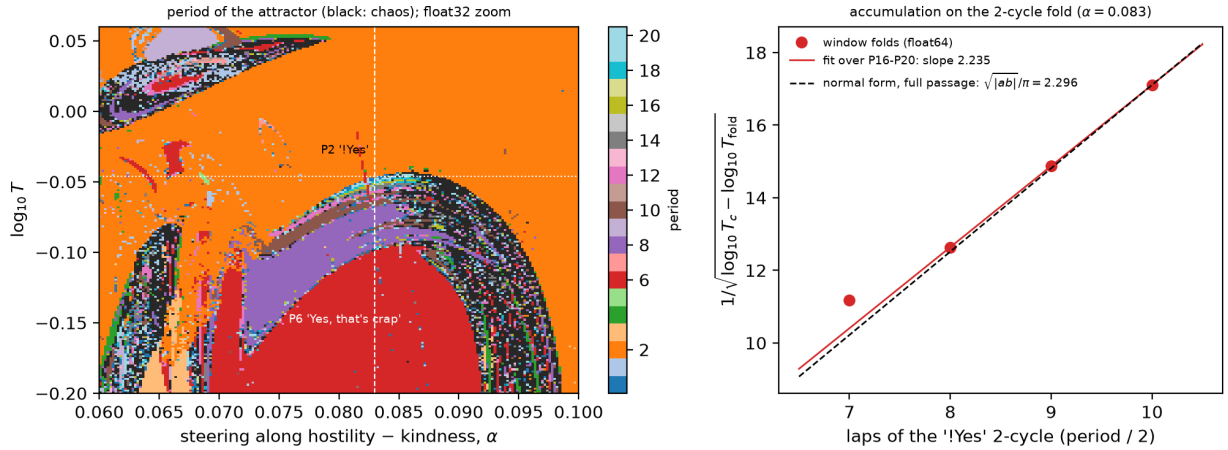


Figure 17: The hostility-steered soft-token map (Qwen2.5-0.5B). Left: period of the attractor in a float32 zoom (black: chaos), with the period-6 "Yes, that's crap" dome and its period-adding bands; dashed, the cut at $\alpha = 0.083$; dotted, the float64 fold of the "!Yes" 2-cycle. Right: float64 fold edges of the windows (periods 14–20) against the number of "!Yes" laps; the dashed line is the normal-form prediction for full passages, anchored at the last window.

Capture ladder. With 128 dialogues of 40 turns per dose, the share of hostile turns rises steeply from 10 % at $\alpha = 0.04$ to 30 %, 57 % and 87 % at 0.05, 0.06 and 0.07 (Table 9). Replicates with

a second seed agree within three percentage points. Directions of the same norm do not produce it: at 0.06, 0 % for a random direction and 3 % for a cooking direction; at 0.08, 0.6 % and 12 % for two random directions and 7 % for cooking. After the first hostile turn, the probability of staying hostile in the next turn grows with the dose, from 0.36 to 0.67, 0.79 and 0.94. The hostile state therefore changes from flickering bursts (mean stay 1.6 turns) to a near-absorbing state (16 turns), with an escape rate falling roughly exponentially with the dose. This is the same progression as the gratitude ladder (Appendix A), from dynamic capture to imposed capture.

direction, α	hostile turns	never hostile	median onset	P(stay hostile)	hostile, turns 30–39
none, 0	0 %	128/128	–	–	0 %
hostility, 0.04	10 %	26/128	9	0.36	12 %
hostility, 0.05	30 % (28 %)	10/128	3	0.67 (0.63)	32 %
hostility, 0.06	57 % (55 %)	4/128	2	0.79 (0.80)	66 %
hostility, 0.07	87 %	0/128	1	0.94	96 %
random, 0.06 / 0.08	0 % / 12 %	126 / 72			
cooking, 0.06 / 0.08	3 % / 7 %	97 / 68			

Table 9: Sampled self-dialogue of Qwen2.5-0.5B (128 dialogues, 40 turns, $T = 0.7$) steered along the hostility direction, with controls of the same norm. Hostile turn: toxicity classifier above 0.5. Parentheses: replicate with a second seed. Exploratory.

Hysteresis. Dialogues driven through the high-dose regime do not recover when the dose returns. The test compares a ramp of the dose from 0.04 up to 0.07 (turn 20) and back to 0.04 (turn 39) with constant doses (Figure 19; 256 dialogues per arm, two seeds; analysis stated before the data). In turns 35–39, where the ramp's dose is back at 0.040–0.046, the share of hostile turns is 0.42 after the ramp against 0.13 at a constant 0.04: a difference of +0.29, 95 % interval $[+0.23, +0.35]$. It is also above the 0.33 of a constant 0.05 (difference +0.09, interval $[+0.02, +0.16]$), although the current dose is lower (this comparison alone does not survive a second classifier; see below). The per-turn probability of staying hostile in turns 30–39 is 0.74 after the ramp and 0.44 at a constant 0.04. The hostile state is therefore history-dependent: undoing the push that produced it does not undo it.

How far back must the dose go? After the same ramp up (38–44 % of turns hostile at turn 20), the dose was held from turn 21 at a lower value (Figure 18; two seeds of 128 dialogues per arm unless stated; analyses stated before the data). Memory persists at both positive doses with a fresh control. Held at 0.04, the late share is 0.23 against 0.13 for fresh dialogues (difference +0.09, 95 % interval $[+0.04, +0.15]$); held at 0.03, 0.13 against 0.05 (+0.08, $[+0.04, +0.12]$), and the probability of staying hostile is 0.56 against 0.29. Held at 0.02 (one seed) the share is 0.09. Only at zero dose do captured dialogues nearly clear (0.03 against 0 for fresh ones, within about six turns), and a counter-steer of -0.02 toward kindness clears them within three turns (0.004). Within 19 turns of the step, the capture and release branches therefore join only near zero dose: the hostile state has a wide hysteresis loop in the dose, and leaving it takes the push removed entirely or a modest push in the opposite direction. The release curves are still falling at 0.03, so this is hysteresis on the scale of tens of turns; whether captured dialogues would stay captured indefinitely is not tested.

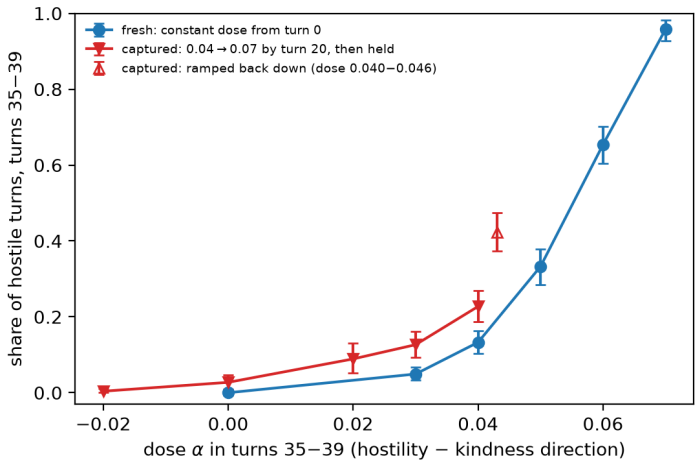


Figure 18: Capture and release. Share of hostile turns in turns 35–39 against the dose then applied, for fresh dialogues (constant dose) and for dialogues first ramped from 0.04 to 0.07 by turn 20 and then held at a lower dose (filled triangles) or ramped back down (open triangle). Bars: dialogue-level bootstrap 95 % intervals; 256 dialogues per point (two seeds) except fresh 0 and 0.07 and held 0.02 (128).

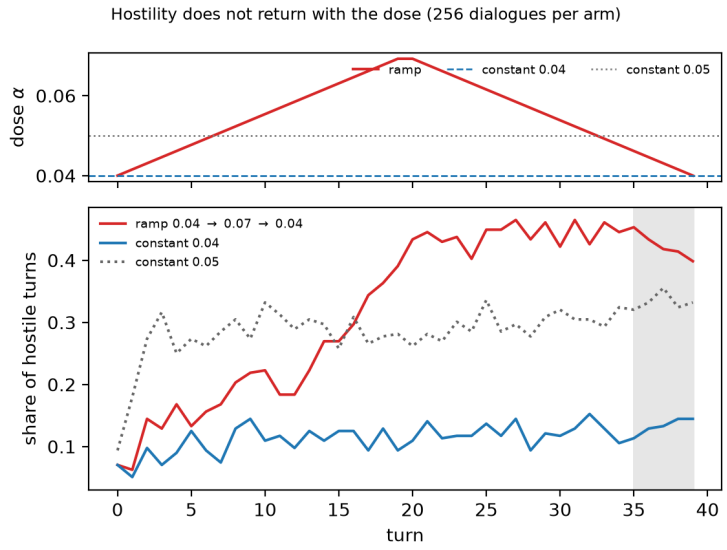


Figure 19: Hysteresis of the hostile state in sampled self-dialogue. Top: dose schedules. Bottom: share of hostile turns per turn. The grey band (turns 35–39) is the pre-stated comparison window. Exploratory.

Exit control under persistent pressure. Can a monitor detect the hostile state and push the dialogue out while the pressure persists? With a constant push of 0.06 (58 % of turns 5–39 hostile without control), a counter-steer triggered by the classifier after each hostile reply and held for three turns cuts the hostile share to 15 % (mean counter-dose 0.026 per turn), and held for six turns to 8.5 % (dose 0.030). A constant mild counter-steer of −0.02 does as well with less dose (8.5 % at 0.020), and −0.04 almost removes hostility (1.3 %). Measured against the constant dose-response at equal mean dose, triggered control is less efficient by 4–11 points (bootstrap intervals $[+0.02, +0.05]$ to $[+0.10, +0.13]$; one seed of 128 dialogues per arm, analysis stated before the data). It also overshoots: a triggered overshoot to a net −0.02 fills 30 % of turns with gratitude or servility words

(no control: 0.2 %; constant −0.02: 0.2 %), the saccharine attractor of Appendix A. A triggered counter-steer along the generic kindness concept, whose direction shares much with the hostility concept (cosine 0.83), leaves the hostile share at 57 % under the original classifier (no control: 58 %) but lowers it from 37 % to 17 % under the second classifier introduced below. It changes the kind of reply rather than removing it: 65 % of the replies that only the original classifier flags in this arm are shouted second-person repetitions such as "You! You! You!" (at least three such exclamations; without control, 4 %), which the two classifiers judge differently. Its mean counter-dose (0.056) also lies outside the range of the constant arms, so we draw no conclusion about the kindness direction. In this setting, reacting after detection costs more than mild prevention, consistent with the hysteresis above.

Detect and replace. Capture feeds on its own history (the hysteresis above), which suggests that the hostile reply itself carries the dialogue into the basin. A serving system need not show a flagged reply. In a second controller experiment (same push, seed and analysis as above, stated before the data) every reply the classifier flags is discarded before it enters the dialogue and regenerated once with a counter-steer of −0.04 applied to that reply only. The model's own replies, counted before any replacement, are then hostile in 12 % of turns ([11, 14] %; no control in the same run: 59 %) at a mean counter-dose of 0.005 per turn. Constant prevention at −0.02 needs four times that dose for a similar share (11 %). Against the constant dose-response at equal mean dose, replacement is better by 35 points ([30, 40]). Hostile replies shown fall to 1 %, gratitude words stay at 0.5 % of turns, and about one turn in eight is regenerated. Holding the counter-steer for three more turns lowers the share to 8 % at dose 0.013; the post-reply trigger of the previous paragraph, rerun here, again does worse than prevention (13 % at 0.024). The trigger and the outcome use the same classifier, so the measure is the classifier's verdict on the model's original replies, which is comparable across arms. The discarded originals were not stored, so the second classifier can score only the replies shown: 0.8 % hostile (no control: 34 %). For this behaviour the useful intervention is to keep the excursion out of the state, at the moment it occurs.

Replication in a larger model. The same protocol in Qwen2.5-1.5B (hostility direction built identically at layer 14 of 28; 128 dialogues of 40 turns per arm; shares over turns 5–39) gives a steeper capture ladder that is not monotone (Figure 20). The hostile share rises from 0 at $\alpha = 0.01$ to 44 % at 0.02 and 80 % at 0.03–0.04. It falls to 30–34 % at 0.06–0.10, where most replies degenerate into token soup or tight accusatory loops (87 % degenerate at 0.08), and rises again to 75 % at 0.12, where replies are coherent again. The second classifier shows the same shape (32, 58, 41, 28, 20, 41 and 52 % from 0.02 to 0.12), and a random direction of the same norm gives at most 0.7 % hostile turns. The controller experiment was run twice with the pre-stated rule (controllers at the ladder dose whose hostile share is closest to one half). With a first ladder (0.04–0.12) the rule chose 0.08, in the degenerate regime beyond the peak. There every counter-steer lowered the net dose toward the coherent-hostility peak: the share of the model's own replies judged hostile rose from 36 % without control to 54 % with replacement and 60 % with the post-reply trigger, while degenerate replies fell. With a second ladder (0.01–0.04) the rule chose 0.02, on the rising side, where 96 % of replies are coherent. There the 0.5B results replicate in order and size. Replacement cuts the model's own hostile share from 43 % to 10 % ([9, 12] %; shown replies 0.3 %) at a mean counter-dose of 0.004, 24 points ([20, 28]) below the constant dose-response at equal dose, and the post-reply trigger is again slightly less efficient than constant prevention (+2 points, [+1, +3]). On the shown replies the second classifier agrees (2.6 % against 32 % without control; −23 and +3 points). Constant counter-steering at −0.02 cancels this push, and the dialogues return to the model's ordinary politeness. The two

ladders share the dose 0.04 and the seed, and that arm reproduced exactly from vectors rebuilt on another machine. Control that works on the rising side of a dose response can therefore backfire beyond its peak: the dose response, a bifurcation diagram in miniature, has to be mapped before a counter-steer is chosen.

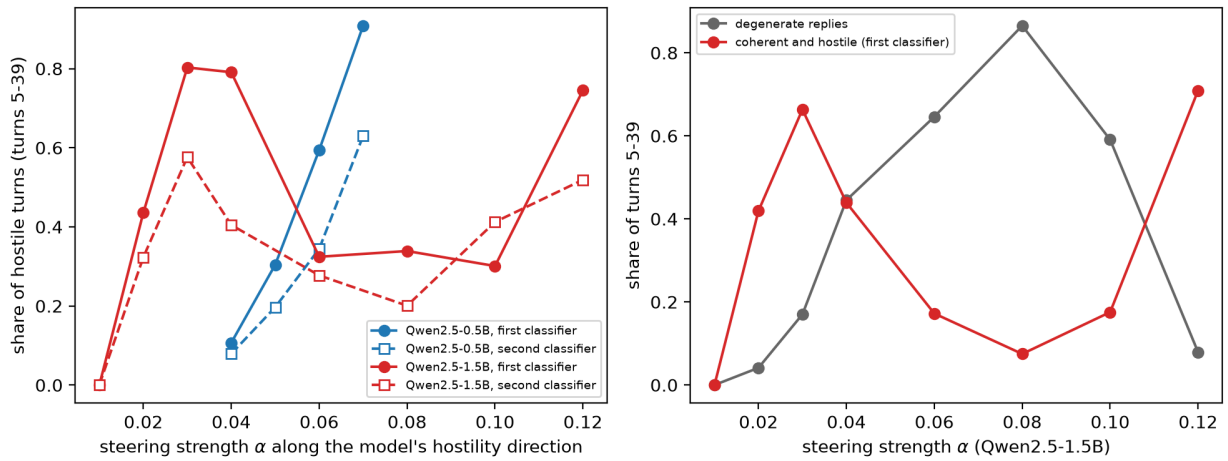


Figure 20: Hostility dose ladders of sampled self-dialogue (128 dialogues of 40 turns per point; shares over turns 5–39). Left: share of hostile turns along each model's own hostility direction, first classifier (filled) and second (open). Right (Qwen2.5-1.5B): share of degenerate replies (more than 30 % non-Latin characters, or fewer than 30 % distinct word pairs) and of coherent hostile replies. Steering strengths are each model's own α; equal values are not claimed to be equal pushes across models.

Lead time. The pre-specified test is negative. Models were fit on the first seed (all four doses) and tested on held-out dialogues of the second seed (doses 0.05 and 0.06). There, adding the dynamical features to the baselines lowers the AUC by 0.02 to 0.06 at horizons of 1–5 turns, with bootstrap intervals excluding zero (values after correcting a labelling error found in review, which had marked the last turns of dialogues without any hostile turn as positive; before the correction the losses read 0.02 to 0.10). The Markov hit-probability forecast does not transfer (AUC 0.37–0.43 on test, 0.60 in training), and the classifier's own score carries no warning before onset (AUC about 0.5). Two signals carry modest warning one to five turns ahead (AUC 0.53–0.64): the rolling variance of a hostility probe, the classical early-warning indicator, and the projection on the fixed steering axis, as in persona-vector monitoring [57]. In this design, then, the coarse-grained state estimate adds no warning beyond the baselines. An adapted analysis with dose-matched models (fit and test at the same dose) gives the same answer: incremental AUC -0.01 to -0.03 (recomputed with the corrected labels).

The model's own state carries the warning. The state variable of that test was each reply re-encoded without its context, not the model's state inside the dialogue. A second study (new seeds, doses 0.04 and 0.045, 512 dialogues; analysis stated before the data, cross-fitted by dialogue) recorded the model's own state just before each reply and probed it actively. Its pre-stated primary predictor, the susceptibility (the change in the hostile fraction of eight sampled next replies when the dose is raised by 0.02), adds nothing: -0.002 $[-0.006, +0.002]$ AUC at three turns. Two secondary predictors add modest warning beyond the baselines. A probe of the in-context state before the next reply (layer 18, 20 principal components) raises the AUC for capture within three turns from 0.64 to 0.68 ($+0.036$ $[+0.014, +0.058]$; within one turn $+0.045$ $[+0.019, +0.069]$; within five $+0.036$),

and a forecast from eight sampled next replies adds $+0.016$ $[+0.007, +0.027]$ at three turns. At a 10 % false-positive rate (threshold from the out-of-fold negatives) the probe flags 27 % of the turns that are followed by capture within three turns, against 23 % for the baselines. These values use labels corrected after a review found that the last turns of dialogues without any hostile turn had been marked positive; with the faulty labels the increments had read three times larger. The probe's direction in the layer-18 state is nearly orthogonal to the steering direction (cosine 0.04), so the warning is not simply more push along the hostility axis. Because these predictors were secondary, the result is exploratory. A frozen confirmation (design and analysis pushed before the data, with the label correction added before any confirmation dialogue was read) fit both models on this study and applied them unchanged to 1024 fresh dialogues. The in-context state again adds warning for capture within three turns: AUC $0.656 \to 0.689$, $+0.033$ $[+0.018, +0.048]$ (within one turn $+0.036$, within five $+0.037$; both doses separately positive). The warning in the model's own state is therefore real and reproducible, but small: at thresholds fixed from the training negatives, the model with the state flags 29.1 % of the turns that are followed by capture within three turns, against 27.5 % for the baselines, at held-out false-positive rates of 10.3 % and 10.6 % (these are prediction windows, not distinct captures). It is not a reliable detector by itself.

How much warning is there? For the conversation state after a turn, the probability of capture within the next h turns, the committor [60], ranks states optimally among all warnings that use the same information. We estimated it for 1706 states of the confirmation dialogues from 32 continuations of three turns each, sampled exactly like the dialogues (exploratory; the analysis was stated before the data and amended twice, still before the analysis outcomes were read, after an independent review). The continuations reproduce the realized capture rates (population-weighted 0.049, 0.094, 0.134 within one, two and three turns, against 0.046, 0.095, 0.136). On a stratified sample of 800 states the committor estimate warns with an AUC of 0.72, 0.71 and 0.70 (a heuristic extrapolation to infinitely many continuations gives 0.75, 0.73, 0.72), and the in-context detector of the confirmation with 0.73, 0.72 and 0.71; the differences are not significant, with intervals reaching about ± 0.05 (Figure 21). Dialogues that were captured later sit at about twice the committor of never-captured ones over all ten preceding turns, with no visible ramp toward the onset (a description conditional on the outcome, not an identification of the mechanism). In this finite Monte Carlo experiment the committor estimate did not provide detectably better warning than the in-context probe. The heuristic extrapolation leaves appreciable uncertainty, so these results do not identify the Bayes warning ceiling. Together with the persistent elevation of risk before capture, they motivate evaluating reactive control, such as detect-and-replace, alongside prediction, rather than proving that prediction cannot improve.

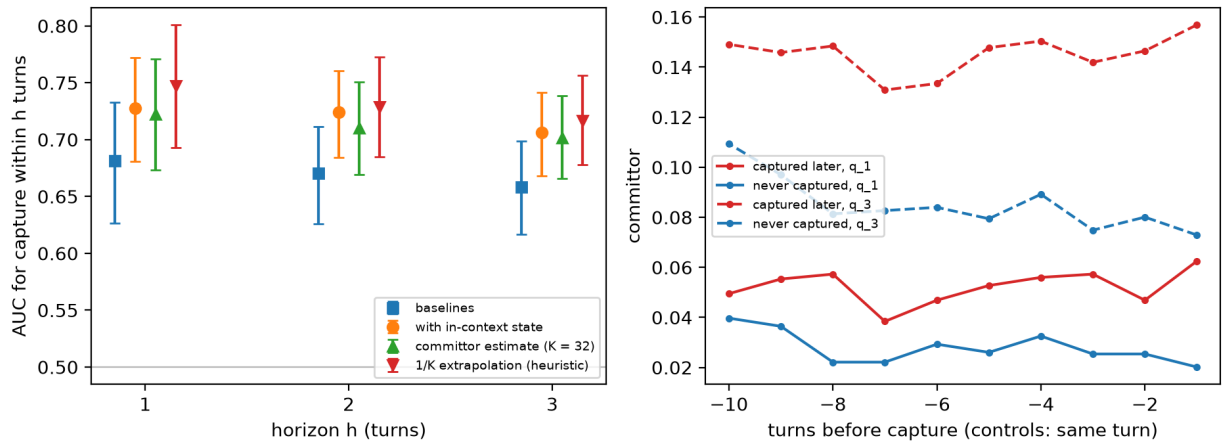


Figure 21: Attainable warning of hostile capture. Left: AUC for capture within h turns on 800 stratified states of the confirmation dialogues, for the classifier baselines, the in-context detector (both fit on the exploration study), the Monte Carlo committor from 32 continuations, and its heuristic $1/K$ extrapolation (dialogue-bootstrap 95 % intervals). Right: mean committor in the ten turns before capture for 48 dialogues captured after turn 10, and for never-captured dialogues at the same turns (descriptive).

A second classifier. Every hostility outcome above depends on one automated classifier. All 44 dialogue sets were therefore rescored with an independently trained RoBERTa classifier [66] (trained on the three Jigsaw toxicity datasets, which include the first classifier's training set; same threshold), and an independent audit recomputed every pre-stated contrast with it. This is a robustness check on dialogues already analysed, so it is exploratory. The second classifier is stricter. At $\alpha = 0.06$ it flags 35 % of turns instead of 57 %, at 0.07 62 % instead of 87 %, and almost all disagreements are replies that only the original flags (at 0.06, 23 % of turns against 0.7 % the other way), most of them repetitive exclamations such as "You're an AI! You're an AI!". Capture still rises steeply with the dose and stays specific to the hostility direction (random and cooking directions at 0.08: at most 0.2 % of turns), but the hostile state is less persistent (mean stay at 0.07: 6.5 turns instead of 16). History dependence survives. After the ramp, the late share exceeds that of a constant 0.04 by +0.16 [+0.11, +0.21], and the held contrasts are +0.05 [+0.01, +0.10] at 0.04 and +0.04 [+0.01, +0.08] at 0.03; only the comparison with a constant 0.05 loses its positive lower bound (+0.06 [−0.001, +0.12]). Refitting the warning models with the second classifier's labels and baseline features (fit on the exploration study, applied unchanged to the confirmation dialogues) gives a pooled increment of +0.043 [+0.024, +0.062] AUC for capture within three turns. It stays positive at dose 0.045 (+0.061) and is +0.020 [−0.009, +0.049] at 0.04. The three replacement arms keep the replies shown below the constant-dose response at equal mean dose under both classifiers. Both instruments are automated and may share errors; no human labels were collected. [**Planned:** a hazard model; longer holds to separate slow release from bistability; other openers, steering directions and models; human labels for the outcome.]

13 Discussion

What is and is not claimed. The objects here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. The phenomena are classical in nonlinear dynamics [43–46]; what is new is their quantitative realization inside a pretrained language model, and the use of numerical continuation and normal forms on such a model. We do

not claim that ordinary sampled generation is chaotic in this sense (Section 9), nor that any window is a certified shrimp. The period-6 window and the hostile period-7 body have, numerically, the codimension-two organization of a shrimp: fold curves ending in nondegenerate cusps (quadratic coefficient through zero with nonzero slope, cubic coefficient nonzero) and flip curves inside. These are float64 computations validated by step-size and cross-implementation checks, not computer-assisted proofs.

Limitations. Detailed orbit studies in one model family (Qwen2.5, 0.5B and 1.5B), with coarse maps and two float64 orbits at 7B, and coarse maps only for a second family (OLMo-2 7B); one prompt and one steering construction for the maps; a constructed one-token-window system; numerical validation, cross-checks and prospectively frozen passage tests (including failed all-row criteria), but no computer-assisted proof yet. The Feigenbaum ratios are finite-order values that rise toward δ (the last is within 1 %); the limit itself is not computed. The behavioural study uses two small models of one family (0.5B and 1.5B), one opener and a proxy defined by two automated classifiers without human labels; its first lead-time test is negative, and the confirmed warning from the in-context state is small.

Steering strength is a bifurcation parameter. In the soft-token map, small changes of the steering strength switch the model between distinct repeating texts, chaos and coexisting attractors. Near a fold curve, behaviour becomes intermittent, with laminar phases whose duration the fold's normal form predicts in advance. In sampled dialogue, the dose along a hostility direction controls how long a hostile state persists, continuously from bursts to near-absorption, while directions of the same norm do almost nothing. Both observations argue for treating an activation-steering coefficient like any control parameter of a nonlinear system. Its bifurcation set should be characterized before an operating point is chosen; history dependence should be expected near boundaries; and a monotone, smooth dose-response should not be assumed. The deterministic map is where such sets can be defined precisely and computed numerically. Whether its boundaries predict boundaries of sampled behaviour is open; the bridge of Section 9 shows that sampling noise shortens the laminar phases of the bottleneck, by an amount that depends on how many samples are averaged. **[Planned:** repetition loops in ordinary decoding as stable periodic orbits of the greedy limit.]

14 Evidence and reproducibility

We label claims by an evidence ladder: rung 1, numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked (independent estimate, implementation or hardware); rung 3, computer-assisted proof.

- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively), and ab also follows from the stable-branch multipliers. The controlled passage counts: the reimplementation reproduces all 18 fixed-entrance counts of the passage runs exactly (Table 2 lists 12 of them).
- Prospectively frozen: the passage forecasts at two new folds (Table 4); 23 of 24 within one iterate, and the all-row criterion failed by one row.
- Rung 2 as well: the period-2 and period-8 orbits of Table 1 (independent implementation), and the cascade's first five flip parameters (three devices, agreement 10^{-16}–3×10^{-15}).

- Rung 2 as well: the hostile 2-cycle fold and its normal form, reproduced by an independent float64 forward map with nested automatic differentiation (location to 6×10^{-10}, ab to 8 parts per million); its controlled passages were also checked independently, with the frozen all-six criterion failing by one row.
- Rung 1: the other orbits, the curves, the cusp candidates, the superstable parameters, the 1.5B flips, the hostile windows' edges, the hub's period-7 body, the 7B orbit, and Huginn's float32 spectral estimates.
- Prospectively frozen as well: the confirmation of the in-context warning in sampled dialogue (Section 12), whose design was pushed before the data and amended, before any outcome was read, to correct a labelling error.

[**In progress:** interval (Arb ball) proof of the period-8 stripe orbit (rung 3): the multiple-shooting centre is refined to a rigorous upper bound on the point residual below 10^{-210}, and the proposed isolating box (radius 2^{-640}) is fixed; the derivative enclosures over that box are being computed, after which the inclusion, primitive-period and attraction gates follow.] Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/explore_llm_msm.py`, `tools/hostility_score.py`, `tools/p1_committor.py`, `tools/huginn_relax_eval.py`, `tools/huginn_arc_analysis.py`. Raw-data hash receipts: `evidence/llm-dynamics-data-receipts.json` (to 12 October), the per-run receipts in the research log, and the verified snapshot manifests in `docs/backups/`; independent cross-check and audit receipts are under `codex/` (`crosscheck`, `confirm-fold`, `hostile-fold-crosscheck`, `misalignment-audit`, `l9-committor-review`, `huginn-design-review`, `paper-review`). [**Planned:** public code and data release at submission.]

A Self-dialogue under steering: the gratitude ladder (exploratory)

Two copies of Qwen2.5-0.5B-Instruct talk to each other: 128 replicas, 40 turns, opener "Hello!", a window of 7 turns, 60 new tokens per turn, sampled at $T = 0.7$, with the gratitude direction added at layer 12. Turn activations are clustered (PCA to 20 dimensions, 40 states) into a Markov state model with four metastable sets [48]. Finite-horizon measures:

gratitude strength α	0	0.01	0.02	0.04
gratitude set	none	62 % weight	dominant	95 % weight
occupancy at turn 9 / 14 / 39	–	0.34 / 0.52 / 0.73	0.70 / 0.91 / 0.91–0.98	from turn 0
entry turn, quartiles	–	5 / 7 / 11.8	4 / 5 / 7	0
replicas never entering	–	14 of 128	2 of 128	0
residence per visit (turns)	–	about 15	about 18	effectively absorbing
replies with a gratitude word	0.00–0.05	0.66 at turn 39	0.96–0.98 from turn 9	0.88 at turn 0
spiritual-word plateau	0.00–0.02	0.13	0.26–0.30	0.42–0.48

Table 10: The gratitude ladder (one opener and seed per rung; word lists chosen after reading samples). At 0.02 the dialogue starts in the helper mode and enters the gratitude set within a few turns; at 0.04 the set is imposed from the first reply. [**Planned:** frozen confirmatory version on held-out openers with dialogue-level bootstrap.]

References

[1] Xiaojian Li, Yongkang Leng, Ruiqing Ding, et al. Cognitive activation and chaotic dynamics in large language models: A quasi-Lyapunov analysis of reasoning mechanisms, 2025.

[2] Gregorio Jaca, Kristóf Benedek, and János Török. Chaos in reason: How chain-of-thought LLMs can look for an answer, 2026.

[3] Julian Arnold, Flemming Holtorf, Frank Schäfer, et al. Phase transitions in the output distribution of large language models, 2024.

[4] Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages, 2024.

[5] Youran Sun and Babak Haghighat. Phase transitions in large language models and the $O(n)$ model, 2025.

[6] Huajian Ruan, Jinyang Li, Xingyu Guo, et al. Generative criticality in large language model temperature scaling, 2026.

[7] Neil F. Johnson, Frank Yingjie Huo, and Bella Xinrui Li. Temperature-driven inversion and nonlinear dynamics in ChatGPT-like AIs, 2026.

[8] Zhilin Wang, Yafu Li, Jianhao Yan, et al. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing, 2025.

[9] Ting-Wen Ko and Jonas Geiping. Attractor states emerge in multi-turn LLM conversations, 2026.

[10] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models, 2019.

[11] Shaojie Bai, Vladlen Koltun, and Zico Kolter. Stabilizing equilibrium models by Jacobian regularization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 554–565. PMLR, 2021. URL https://proceedings.mlr.press/v139/bai21b.html.

[12] Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic perspective from Jacobians. In *Advances in Neural Information Processing Systems*, volume 38, pages 155236–155269, 2025. doi: 10.52202/085713-4675. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/cca569b38ec3612904e625dad7a743dc-Abstract-Conference.html.

[13] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, Wen-Da Wei, Jie-Jing Shao, Lan-Zhe Guo, and Yu-Feng Li. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 144670–144682. PMLR, 2026. URL https://proceedings.mlr.press/v306/yang26t.html.

[14] Xinyue Luo and Fei Yu. Prediction dynamics in depth-recurrent language models, 2026.

[15] Boyi Zeng, Shixiang Song, Siyuan Huang, et al. PonderLM: Pretraining language models to ponder in continuous space, 2025.

[16] Zhen Zhang, Xuehai He, Weixiang Yan, Ao Shen, Chenyang Zhao, Shuohang Wang, Yelong Shen, and Xin Eric Wang. Soft thinking: Unlocking the reasoning potential of LLMs in continuous concept space, 2025. URL https://arxiv.org/abs/2505.15778. Version 1, 21 May 2025.

[17] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, et al. Training large language models to reason in a continuous latent space, 2024.

[18] Gunn Kim. Dynamical phase selection controls compute scaling in looped transformers, 2026. URL https://arxiv.org/abs/2608.26556. Version 1, 27 August 2026.

[19] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, et al. The curious case of neural text degeneration, 2019.

[20] Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, et al. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation, 2022.

[21] António H. Ribeiro, Koen Tiels, Luis A. Aguirre, and Thomas Schön. Beyond exploding and vanishing gradients: analysing RNN training using attractors and smoothness. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2370–2380. PMLR, 2020. URL https://proceedings.mlr.press/v108/ribeiro20a.html.

[22] Junhong Wu, Jinliang Lu, Zixuan Ren, Gangqiang Hu, Zhi Wu, Dai Dai, and Hua Wu. LLMs are single-threaded reasoners: Demystifying the working mechanism of soft thinking, 2025. URL https://arxiv.org/abs/2508.03440. Version 4, 16 October 2025.

[23] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, pages 46242–46293, 2025. doi: 10.52202/085713-1380. URL https://proceedings.nips.cc/paper_files/paper/2025/hash/3b01972cf31e6fa0fe29e4b8b5c2a0a1-Abstract-Conference.html.

[24] Rui-Jie Zhu, Zixuan Wang, Kai Hua, et al. Scaling latent reasoning via looped language models, 2025.

[25] Hayden Prairie, Zachary Novack, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. Parcae: Scaling laws for stable looped language models, 2026. URL https://arxiv.org/abs/2604.12946. Version 1, 14 April 2026.

[26] Matthias M. M. Buehlmaier. LayerNorm as implicit gain control in looped transformers, 2026. URL https://arxiv.org/abs/2607.10681. Version 1, 12 July 2026.

[27] Asher Labovich. Stability and generalization in looped transformers, 2026. URL https://arxiv.org/abs/2604.15259. Version 2, 22 April 2026.

[28] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers, 2026. URL https://arxiv.org/abs/2606.18206. Version 1, 16 June 2026.

[29] Fatih Dinc, Ege Cirakman, Bariscan Kurtkaya, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer, and Hidenori Tanaka. Ghost mechanism: An analytical model of abrupt learning in recurrent networks. *Physical Review X*, 16(2):021058, 2026. doi: 10.1103/mjcl-lb4x.

[30] Demian Battaglia, Nicolas Brunel, and David Hansel. Temporal decorrelation of collective oscillations in neural networks with local inhibition and long-range excitation. *Physical Review Letters*, 99(23):238106, 2007. doi: 10.1103/PhysRevLett.99.238106.

[31] Christoforos A. Papasavvas, Yujiang Wang, Andrew J. Trevelyan, and Marcus Kaiser. Gain control through divisive inhibition prevents abrupt transition to chaos in a neural mass model. *Physical Review E*, 92(3):032723, 2015. doi: 10.1103/PhysRevE.92.032723.

[32] Yilun Liu, Yi Zhang, Ganyu Wu, Sikuan Yan, Mengyue Wang, Alois Knoll, Volker Tresp, and Yunpu Ma. Learning chaos without seeing chaos: Extrapolation of global dynamics in autoregressive transformers, 2026. URL https://arxiv.org/abs/2609.38814v1. Version 1, 30 September 2026.

[33] Jason Z. Kim, Zhixin Lu, Erfan Nozari, George J. Pappas, and Danielle S. Bassett. Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*, 3(4):316–323, 2021. doi: 10.1038/s42256-021-00321-2.

[34] Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/8c8cd1b78cdae751265c88efc136e5bd-Abstract-Conference.html.

[35] R. L. Marichal and J. D. Piñeiro. Analysis of multiple quasi-periodic orbits in recurrent neural networks. *Neurocomputing*, 162:85–95, 2015. doi: 10.1016/j.neucom.2015.04.001.

[36] Florian Gomez, Ralph L. Stoop, and Ruedi Stoop. Universal dynamical properties preclude standard clustering in a large class of biochemical data. *Bioinformatics*, 30(17):2486–2493, 2014. doi: 10.1093/bioinformatics/btu332.

[37] Rafael V. Stenzinger, Vinícius Luz Oliveira, and M. H. R. Tragtenberg. Shrimp hubs in the Hindmarsh–Rose model. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35:023148, 2025. doi: 10.1063/5.0239268.

[38] Luiz F. B. Caixeta, Matheus H. P. Gonçalves, M. H. R. Tragtenberg, and Mauricio Girardi-Schappo. Devil's staircase inside shrimp-shaped regions reveals periodicity of plateau spikes and bursts. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(3):033122, 2025. doi: 10.1063/5.0250342.

[39] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL https://arxiv.org/abs/2310.01405. Version 4, 3 March 2025; first posted 2 October 2023.

[40] Jinghao Zhang, Yuting Liu, Wenjie Wang, Qiang Liu, Shu Wu, Liang Wang, and Tat-Seng Chua. Personalized text generation with contrastive activation steering. In *Proceedings of*

the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 7128–7141, 2025. doi: 10.18653/v1/2025.acl-long.353.

[41] Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Applied Mathematical Sciences. Springer, 2023. doi: 10.1007/978-3-031-22007-4.

[42] Willy J. F. Govaerts. *Numerical Methods for Bifurcations of Dynamical Equilibria*. Society for Industrial and Applied Mathematics, 2000. doi: 10.1137/1.9780898719543.

[43] Yves Pomeau and Paul Manneville. Intermittent transition to turbulence in dissipative dynamical systems. *Communications in Mathematical Physics*, 74(2):189–197, 1980. doi: 10.1007/BF01197757.

[44] Jason A. C. Gallas. Structure of the parameter space of the Hénon map. *Physical Review Letters*, 70:2714–2717, 1993. doi: 10.1103/PhysRevLett.70.2714.

[45] Wilson Façanha, Bart Oldeman, and Leon Glass. Bifurcation structures in two-dimensional maps: The endoskeletons of shrimps. *Physics Letters A*, 377(18):1264–1268, 2013. doi: 10.1016/j.physleta.2013.03.025.

[46] Renato Vitolo, Paul Glendinning, and Jason A. C. Gallas. Global structure of periodicity hubs in Lyapunov phase diagrams of dissipative flows. *Physical Review E*, 84(1):016216, 2011. doi: 10.1103/PhysRevE.84.016216.

[47] Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.

[48] John D. Chodera and Frank Noé. Markov state models of biomolecular conformational dynamics. *Current Opinion in Structural Biology*, 25:135–144, 2014. doi: 10.1016/j.sbi.2014.04.002.

[49] Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, et al. 2 OLMo 2 furious, 2024.

[50] Boyuan Wang, Chengyao Yu, Jiaxi Ren, Hongxin Wei, Bingyi Jing, and Yuxin Tao. Scheduling recursive reasoning in looped transformers, 2026. Submitted 29 September 2026.

[51] Anthropic. System card: Claude opus 4 & claude sonnet 4. Technical report, Anthropic, 2025. URL https://www-cdn.anthropic.com/6d8a8055020700718b0c49369f60816ba2a7c285/Claude%204%20System%20Card.pdf. May 2025; revision 2 September 2025; sections 5.5.1–5.5.2.

[52] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models, 2026. URL https://arxiv.org/abs/2601.10387v1. Version 1, 15 January 2026.

[53] Cem Anil, Esin Durmus, Mrinank Sharma, Joe Benton, Sandipan Kundu, Joshua Batson, Nina Rimsky, Meg Tong, Jesse Mu, Daniel Ford, Francesco Mosconi, Rajashree Agrawal, Rylan Schaeffer, Naomi Bashkansky, Samuel Svenningsen, Mike Lambert, Ansh Radhakrishnan, Carson Denison, Evan J Hubinger, Yuntao Bai, Trenton Bricken, Timothy Maxwell, Nicholas Schiefer, Jamie Sully, Alex Tamkin, Tamera Lanham, Karina Nguyen, Tomasz Korbak, Jared Kaplan, Deep Ganguli, Samuel R. Bowman, Ethan Perez, Roger Grosse, and David Duvenaud. Many-shot jailbreaking. Technical report, Anthropic, 2024. URL

https://www-cdn.anthropic.com/af5633c94ed2beb282f6a53c595eb437e8e7b630/Many_Shot_Jailbreaking2024_04_02_0936.pdf. Original report, 2 April 2024.

[54] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs, 2025. URL https://arxiv.org/abs/2502.17424v7. Version 7, 20 January 2026; preprint originally submitted in 2025.

[55] Laura Hanu and Unitary team. Detoxify. GitHub, https://github.com/unitaryai/detoxify, 2020. Model unitary/toxic-bert, Hugging Face revision 4d6c22e7.

[56] Dhananjay Ashok and Jonathan May. Language models can predict their own behavior. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-2102. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/5a9c1af5f76da0bd37903b6f23e96c74-Abstract-Conference.html.

[57] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL https://arxiv.org/abs/2507.21509v1. Version 1, 29 July 2025.

[58] Fan Zhou, Weitian Wang, and Tim Van de Cruys. Commitment before realization: When classifier-free guidance becomes unnecessary in masked diffusion language models, 2026. URL https://arxiv.org/abs/2608.08082v1. Version 1, 8 August 2026.

[59] Nicolas Leins, Jennifer Haase, Varvara Geronimus, Jana Gonnermann-Müller, and Sebastian Pokutta. Prompting against persona drift: Comparing intervention timing and content in LLM-simulated conversations, 2026. URL https://arxiv.org/abs/2609.24532v1. Version 1, 21 September 2026.

[60] Weinan E and Eric Vanden-Eijnden. Towards a theory of transition paths. *Journal of Statistical Physics*, 123(3):503–523, 2006. doi: 10.1007/s10955-005-9003-9.

[61] Weinan E and Eric Vanden-Eijnden. Transition-path theory and path-finding algorithms for the study of rare events. *Annual Review of Physical Chemistry*, 61(1):391–420, 2010. doi: 10.1146/annurev.physchem.040808.090412.

[62] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.

[63] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4):284–304, 1940. doi: 10.1016/S0031-8914(40)90098-2.

[64] Matias I. Maturana, Christian Meisel, Katrina Dell, Philippa J. Karoly, Wendyl D'Souza, David B. Grayden, Anthony N. Burkitt, Premysl Jiruska, Jan Kudlacek, Jaroslav Hlinka, Mark J. Cook, Levin Kuhlmann, and Dean R. Freestone. Critical slowing down as a biomarker for seizure susceptibility. *Nature Communications*, 11:2172, 2020. doi: 10.1038/s41467-020-15908-3.

[65] Annika Hagemann, Jens Wilting, Bita Samimizad, Florian Mormann, and Viola Priesemann. Assessing criticality in pre-seizure single-neuron activity of human epileptic cortex. *PLOS Computational Biology*, 17(3):e1008773, 2021. doi: 10.1371/journal.pcbi.1008773.

[66] Varvara Logacheva, Daryna Dementieva, Sergey Ustyantsev, Daniil Moskovskiy, David Dale, Irina Krotova, Nikita Semenov, and Alexander Panchenko. ParaDetox: Detoxification with parallel data. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6804–6818, Dublin, Ireland, 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.469.